

A PROXIMAL ORACLE SEQUENTIAL NLP ALGORITHM FOR CONSTRAINED NONSMOOTH OPTIMIZATION

COSMIN G. PETRA*

Abstract. We develop a proximal sequential nonlinear programming (NLP) method for constrained finite-sum optimization with nonsmooth, nonconvex objective terms. Each nonsmooth summand is assumed only to be proper, lower semicontinuous, and prox-bounded, and is accessed through a proximal oracle requiring no derivatives or sensitivity information. The method applies Moreau envelope regularization separately to each summand, preserving separability while retaining the original smooth nonlinear constraints. We establish consistency of this distributed Moreau continuation: as the regularization parameter vanishes, optimal values and global minimizers of the regularized problems converge, along subsequences, to those of the original problem. We then propose a continuation algorithm whose inner loop solves sequential NLP majorization models. These models enforce the smooth constraints exactly and represent the regularized nonsmooth terms by proximal oracle subgradient linearizations plus quadratic stabilization, using the upper- C^2 property of Moreau envelopes; an acceptance-ratio test adaptively updates the stabilization parameter. For fixed regularization, the cluster points of the inner iteration satisfy the stationarity conditions for the regularized problem, and the inner loop terminates finitely for any positive stationarity tolerance. For prescribed positive regularization and stationarity tolerances, the full continuation scheme also terminates finitely and returns an approximately limiting sum-stationary point with an explicit error bound. As the regularization and tolerances vanish, cluster points satisfy first-order limiting stationarity for the original nonsmooth problem under standard constraint qualifications and an attentive convergence condition, which simplify for locally Lipschitz terms. Finally, we show how the algorithm applies when the nonsmooth summands are optimal value functions of parametric optimization problems, for which the proximal oracles can be implemented by quadratically stabilized subproblems.

1. Introduction. We consider optimization problems of the form

$$(1.1) \quad \min_{x \in \mathbb{R}^n} F(x) := f(x) + \sum_{i \in \mathcal{K}} r_i(x),$$

$$(1.2) \quad \text{s.t. } h(x) = 0, \quad x_\ell \leq x \leq x_u,$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ are C^2 , and each $r_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is proper, lower semicontinuous (lsc), finite-valued, and possibly nonconvex and nonsmooth, for each i in a finite index set $\mathcal{K}$. Problems with this structure arise in several applications; in particular, our main motivation comes from stochastic programming with recourse and security-constrained power system optimization, where the functions r_i are optimal value functions of parametric optimization problems, as elaborated in Section 5.

We assume that the functions r_i are accessed through an oracle. In many applications, evaluating $r_i(x)$ already requires solving an optimization subproblem, while derivatives of r_i with respect to x are unavailable, unreliable, or expensive to compute. Rather than requiring such sensitivities, we assume access to a proximal oracle, namely the value of the Moreau envelope [30] of r_i

$$(1.3) \quad e_\lambda r_i(x) := \inf_{z \in \mathbb{R}^n} \left\{ r_i(z) + \frac{1}{2\lambda} \|z - x\|^2 \right\},$$

together with a proximal point

$$w_i \in \text{prox}_\lambda r_i(x) := \arg \min_{z \in \mathbb{R}^n} \left\{ r_i(z) + \frac{1}{2\lambda} \|z - x\|^2 \right\}, \quad \forall i \in \mathcal{K},$$

*Center for Applied Scientific Computing, Lawrence Livermore National Laboratory, Livermore, CA, USA. Email: petra1@llnl.gov. Release number: LLNL-JRNL-2019378

for $\lambda > 0$. When r_i is defined through an optimization problem, existing numerical machinery can often be adapted to evaluate this oracle with little structural change.

Our starting point is the observation that the Moreau envelope regularizes each nonsmooth term. Under the mild assumption of prox-boundedness, $e_\lambda r_i$ converges pointwise to r_i as $\lambda \downarrow 0$ and is locally Lipschitz and upper- C^2 for sufficiently small λ . This suggests replacing (1.1)–(1.2) by the family of regularized problems

$$(1.4) \quad \min_{x \in \mathbb{R}^n} F_\lambda(x) := f(x) + \sum_{i \in \mathcal{K}} e_\lambda r_i(x)$$

$$(1.5) \quad \text{s.t. } h(x) = 0 \text{ and } x_\ell \leq x \leq x_u,$$

and driving $\lambda \downarrow 0$ in a continuation scheme.

For fixed λ , the objective in (1.4) is a sum of a smooth term and upper- C^2 terms, and is therefore upper- C^2 . At first sight, this places the fixed- λ problem within the scope of the sequential quadratic programming (SQP) framework of Wang and Petra [33] for nonsmooth problems with upper- C^2 objectives. The present work, however, is not a direct application of that framework. In [33], the nonsmooth objective (1.4) is treated as an upper- C^2 function for which Clarke subgradients are assumed to be available; here, we do not assume access to subgradients of either r_i or $e_\lambda r_i$. The only information used by the algorithm is a proximal oracle for each r_i .

The use of Moreau envelopes leads to a different stationarity mechanism, which ultimately provides limiting/Mordukhovich subgradients stationarity for the original problem and is stronger than the Clarke stationarity of [33] even for the regularized problems, despite apparently not using subgradients. The key idea is to use proximal points $w_i \in \text{prox}_\lambda r_i(x)$, produced by the oracle, to form the vectors

$$v_i = \frac{1}{\lambda}(x - w_i).$$

For general lsc r_i , the vectors v_i may not be limiting subgradients of $e_\lambda r_i$ at x , but:

- (i) satisfy a majorization bound specific to upper- C^2 functions; and
- (ii) are limiting subgradients of r_i at the nearby proximal point w_i .

The property (ii) is noteworthy and plays an important role in the analysis of the algorithm proposed here. The property stems from the use of Moreau envelopes and is not present for general upper- C^2 functions. We refer to the vectors v_i as *proximal oracle subgradients* to emphasize their Moreau envelope origin and to distinguish them from proximal subgradients previously used in the literature [30].

Consequently, property (i) allows us to develop a sequential proximal oracle NLP algorithm for the regularized, fixed- λ constrained problems (1.4)–(1.5). The method builds convex quadratic stabilized models for the nonsmooth objective term(s) using only proximal points and uses an NLP model for step computations. The choice of an NLP model instead of an SQP model is made to enforce the smooth constraints and objective term exactly, as required in important classes of problems, such as power grid optimization, which motivates this work. An acceptance ratio test is also devised to adjust the stabilization parameter and obtain a simple descent-type of scheme. We prove that the inner, fixed- λ iterations result in cluster points that satisfy stationarity conditions for the regularized problem and terminate finitely for any positive stationarity tolerance.

Furthermore, the property (ii) above yields an outer-loop convergence analysis for the continuation scheme that carries, as $\lambda \downarrow 0$, the stationarity certificates for the fixed- λ problems to limiting subgradients-based stationarity of the original nonsmooth

problem. More specifically, under standard constraint qualifications (CQs) and a mild attentive convergence condition—which holds automatically in several important cases, including locally Lipschitz r_i —we show that every cluster point generated as $\lambda \downarrow 0$ satisfies first-order stationarity conditions for (1.1)–(1.2).

Applying Moreau regularization separately to individual summands r_i preserves the finite-sum structure in the regularized problems, allows parallel proximal computations, and results in single-level decomposition with significant computational parallelism. For example, we have previously shown that this single-level decomposition is particularly well suited when each r_i is the optimal value of a computationally expensive optimization subproblem and can scale up to exascale supercomputers [20]. However, since infimal convolution does not in general commute with summation, replacing each r_i by $e_\lambda r_i$ is not merely a rewriting of the Moreau envelope of the full nonsmooth objective. We show that, under mild assumptions, the regularized problems (1.4)–(1.5) are nevertheless consistent with the original problem, in the sense that global minimizers of the regularized problems converge, along subsequences, to global minimizers of (1.1)–(1.2) as $\lambda \downarrow 0$.

1.1. Related work. Our approach is related to proximal point algorithms (PPAs) [4, 5, 26], but differs in an essential way. Classical PPAs regularize the entire objective and would require evaluating the proximal mapping of F , or equivalently solving subproblems involving $e_\lambda F$. In our setting, this would couple all objective terms and destroy the separable structure and the parallelization potential of the single-level decomposition associated with the nonsmooth part.

Compared with Douglas-Rachford splitting methods [21, 31, 32, 17] and related alternating direction method of multipliers schemes [34, 15, 1, 2, 16], the present framework is designed to work directly with the native structure of (1.1)–(1.2). Splitting methods are typically most natural after reformulating the problem into a two-block or linearly constrained consensus form, which may require aggregating the nonsmooth terms or introducing auxiliary variables and coupling constraints. In contrast, we keep the smooth nonlinear constraints explicit and preserve the finite-sum structure.

The closest broad framework is the constrained composite augmented Lagrangian approach of [12], which treats problems with a single nonsmooth composite term and general set-membership constraints. The framework of [12] is more general, but it does not take advantage of the particular oracle and separability structures that motivate this work. Our setting is deliberately more structured: each nonsmooth summand is accessed through its proximal oracle, and the original smooth nonlinear constraints are retained explicitly rather than absorbed into a reformulated composite constraint, which would obscure the finite sum structure the term-wise subgradients needed in the continuation analysis.

The inner-loop sequential NLP approach here is related to nonsmooth SQP and gradient-sampling methods; see, for example, [35, 10, 6, 8, 9]. These methods build SQP models using sampled gradients or quasi-Newton approximations for locally Lipschitz nonsmooth constrained problems. However, our stationarity analysis relies on proximal points and deterministic nearby limiting subgradients rather than on Clarke subgradient sampling or penalty-SQP stationarity. Moreover, our setting assumes no derivative or sampled-gradient access to the nonsmooth summands.

1.2. Contributions. The paper’s main contribution is a proximal-oracle sequential NLP algorithm for constrained finite-sum nonsmooth, nonconvex optimization under very general assumptions: each nonsmooth summand need only be proper, lsc, and prox-bounded, with no convexity, weak convexity, prox-regularity, or dif-

ferentiability required. The algorithm uses proximal oracle evaluations rather than sensitivity information and preserves the single-level decomposition of the original problem. It provides finite inner-loop termination and stationarity for the regularized problems by using the upper- C^2 property and proximal oracle subgradients features specific to Moreau envelopes. The stationarity certificates from the regularized problems are general enough to be carried via an outer continuation analysis and to obtain limiting first-order sum-stationarity for the original nonsmooth problem under standard CQs and a mild attentive convergence condition. The algorithm applies naturally to and is well suited for nonsmooth terms defined by optimal value functions, where proximal oracles can be implemented via stabilized subproblems.

1.3. Organization. The remainder of the paper is organized as follows. Section 2 reviews the variational analysis tools used throughout, with emphasis on upper- C^2 functions and subdifferential properties of Moreau envelopes. Section 3 establishes the well-posedness and consistency of the Moreau envelope continuation approach as the regularization parameter tends to zero. Section 4 presents the numerical algorithm and analyzes both the fixed- λ inner iterations and the outer continuation scheme. Section 5 discusses how the framework applies naturally to parametric recourse optimization models, with a case study for electric power grids. The focus here is theoretical rather than computational, as the present paper establishes the prerequisites needed to justify and support future computational developments. Section 6 summarizes some of these developments and provides concluding remarks.

2. Preliminary technical results. Let $r : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$, where $\overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$, and let $\bar{x}$ be such $r(\bar{x}) \in \mathbb{R}$. A vector $v \in \mathbb{R}^n$ is in the regular (Fréchet) subdifferential [30] (denoted by $\hat{\partial}r(\bar{x})$) if $r(x) \geq r(\bar{x}) + \langle v, x - \bar{x} \rangle + o(\|x - \bar{x}\|)$.

Following again [30, Definition 8.2], we define the limiting (also known as Morukhovich) subdifferential, denoted by $\partial r(\bar{x})$. A vector $v \in \mathbb{R}^n$ is a limiting subgradient of r at $\bar{x}$ (written $v \in \partial f(\bar{x})$) if there exist sequences

$$x^\nu \rightarrow_r \bar{x} \text{ and } v^\nu \in \hat{\partial}r(x^\nu) \text{ such that } v^\nu \rightarrow v.$$

Here, $x^\nu \rightarrow_r \bar{x}$ denotes the r -attentive convergence and holds whenever $x^\nu \rightarrow \bar{x}$ and $r(x^\nu) \rightarrow r(\bar{x})$ [30, Def. 8.2]. The *horizon subdifferential* is

$$\partial^\infty r(\bar{x}) = \left\{ v^\infty : \exists x^\nu \rightarrow_r \bar{x}, t_\nu \downarrow 0, \text{ and } v^\nu \in \hat{\partial}r(x^\nu) \text{ such that } t^\nu v_\nu \rightarrow v^\infty, \right\}.$$

For a closed set $C \subset \mathbb{R}^n$, the *limiting normal cone* is defined as $N_C(x) := \partial \delta_C(x)$, where δ_C is the indicator of C , defined as $\delta_C(x) = 0$ if $x \in C$ and $+\infty$ otherwise.

For a Lipschitz continuous function r we also use Clarke subdifferentials [30, 29, 7] defined by

$$\partial^C r(x) = \{v \in \mathbb{R}^n \mid r^\circ(x; d) \geq \langle v, d \rangle \text{ for all } d \in \mathbb{R}^n\}.$$

where $r^\circ(x; d)$ is the directional Clarke derivative along direction $d \in \mathbb{R}^n$, defined as

$$r^\circ(x; d) = \limsup_{\substack{y \rightarrow x \\ t \downarrow 0}} \frac{r(y + td) - r(y)}{t}.$$

The function r is called *prox-bounded* if there exists some $\lambda > 0$ such that $e_\lambda r(x) > -\infty$ for some $x \in \mathbb{R}^n$. The prox-threshold of r is the supremum over all such $\lambda > 0$. For lsc, proper functions, prox-boundedness is equivalent to the function $x \mapsto r +$

$\frac{1}{2}\alpha\|\cdot\|^2$ being bounded from below for some $\alpha > 0$ [30, Exercise 1.24]. We remark that any function bounded from below is prox-bounded (with prox-threshold $+\infty$). Also, for every $x \in \mathbb{R}^n$, $\text{prox}_\lambda r(x)$ is nonempty and compact [30, Theorem 1.25] for any $\lambda > 0$ smaller than the prox-threshold.

We build on the concept of upper- C^k ($k \in \{1, 2, \dots\}$) functions in this work. Following Rockafellar and Wets [30, Definition 10.29], a real-valued function f on an open set $O \in \mathbb{R}^n$ is upper- C^k if locally r is the minimum of a family of C^k functions indexed over a compact set, namely, $r(x) = \min_{t \in T} f_t(x)$ for all x in a neighborhood of $\bar{x} \in O$, with T compact, $f_t \in C^k$, and the values and all partial derivatives up to order k depending jointly continuously on (t, x) . The minimum of finitely many C^k functions is therefore upper- C^k , but may fail to be differentiable. There is no distinction for orders $k \geq 2$ since upper- C^2 implies upper- C^∞ [30, Corollary 10.34]. Some prominent members of the upper- C^2 class are squared distance functions to closed sets and Moreau envelopes of lsc, proper, prox-bounded functions for sufficiently small regularization parameters.

A useful equivalent characterization of upper- C^2 functions is via local difference of smooth and finite convex function, namely a function r is upper- C^2 if and only if locally $r = r_{sm} - r_{co}$ with r_{co} a finite convex function and $r_{sm} \in C^2$ [30, Theorem 10.33]. In fact, one may take $r_{sm} = \frac{\rho}{2}\|x\|_2^2$ locally [30, Theorem 10.33], which is equivalent to $r - \frac{\rho}{2}\|\cdot\|_2^2$ being concave locally. Recently, this property has been exploited to devise simplified nonsmooth algorithms for upper- C^2 objectives that need not maintain bundles, *e.g.*, [33], and we also leverage it in this work (see quadratic majorization Lemma 4.3) to obtain convergence to stationary points of the regularized problems (see Proposition 4.5 and Theorem 4.6) using relaxed “proximal” subgradients (see Lemma 2.1 below).

Conversely, a function r is lower- C^k if $-r$ is upper- C^k . Lower- C^2 functions enjoy better regularity than upper- C^2 , in particular they are subdifferentially regular (also known as lower regular) and, for example, the concepts of regular, limiting, and Clarke subdifferentials coincide [30]. On the other hand, the strict inclusion $\hat{\partial} \subset \partial \subset \partial^C$ can hold for upper- C^2 functions in general.

A salient feature of Moreau envelopes of lsc, proper, and prox-bounded functions, not available for general upper- C^2 functions, is that their limiting subdifferentials can be characterized by means of proximal points as shown by the following lemma.

LEMMA 2.1. *Let $r : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ be proper lsc and prox-bounded, and let $\lambda > 0$ be below the prox-threshold. We have the following characterizations of limiting subdifferentials of the Moreau envelopes at a given $x \in \mathbb{R}^n$:*

- (i) $\text{prox}_\lambda r(x) \subset (I + \lambda \partial r)^{-1}(x)$;
- (ii) $\partial e_{\lambda r}(x) \subset \frac{1}{\lambda}(x - \text{prox}_\lambda r(x))$;
- (iii) *For every $v \in \partial e_{\lambda r}(x)$ there exists $y \in \text{prox}_\lambda r(x)$ such that $v = \frac{x-y}{\lambda}$ and $v \in \partial r(y)$.*

Proof. (i) and (ii) are Examples 10.2 and 10.32 from [30], respectively.

(iii) Consider $v \in \partial e_{\lambda r}(x)$. (ii) implies there exists $y \in \text{prox}_\lambda r(x)$ such that $v = \frac{x-y}{\lambda}$. Also, (i) indicates that $x \in y + \lambda \partial r(y)$, which implies $v \in \partial r(y)$ and completes the proof. $\square$

Lemma 2.1 (ii) above shows that any (limiting) subgradient $v \in \partial e_{\lambda r}(x)$ corresponds to a minimizer $y \in \text{prox}_\lambda r(x)$ of the Moreau envelope optimization problem (1.3). However, the converse is not necessarily true as the inclusion in (ii) can be strict. Intuitively, this can occur, for example, when $\text{prox}_\lambda r(x)$ is not a singleton, as

illustrated by the following example.

Example 2.2. Consider $e_{\frac{1}{2}}r(x) = \frac{1}{2}x^2 - |x| + \frac{1}{2}$, the Moreau envelope of $r(z) = \frac{1}{2} - z^2$ if $|z| \leq \frac{1}{2}$ and $r(z) = (|z| - 1)^2$ if $|z| \geq \frac{1}{2}$. For this, we observe that

$$\partial e_{1/2}r(0) = \{-1, 1\} \subsetneq [-1, 1] = 2(0 - \text{prox}_{1/2}r(0)).$$

A complication caused by the possibly strict inclusion in (ii) of Lemma 2.1 is that minimizers/proximal points $y \in \text{prox}_{\lambda}r(x)$ of (1.3) do not necessarily provide a subgradient in $\partial e_{\lambda}r(x)$. In the numerical algorithm proposed in Section 4, we circumvent this limitation by using the proximal oracle subgradients $v = \frac{1}{\lambda}(x - y)$ instead of the true limiting subgradients of $\partial e_{\lambda}r$ at x to solve the Moreau regularized problem for fixed $\lambda > 0$. Notably, Lemma 2.1 (i) shows that these proximal oracle subgradients are limiting subgradients of r at proximal points $y \in \text{prox}_{\lambda}r(x)$. This fact and the convergence consistency between proximal points and algorithm iterates (see Lemma 4.8) are key ingredients in obtaining (asymptotic) convergence of the algorithm to a limiting stationary point to the target problem (1.1)–(1.2) without using the limiting subgradients of the Moreau envelope approximants from (1.4)–(1.5).

Remark 2.3. If r is convex, or more generally weakly convex [11], then for λ small enough, $\text{prox}_{\lambda}r$ is single valued and $e_{\lambda}r$ is C^1 , so Lemma 2.1 (ii) holds with equality. For prox-regular [28] or lower- C^2 functions, same conclusion holds locally on neighborhoods where the proximal mapping is single-valued [30].

In the remainder of the paper, we work under the following two assumptions.

Assumption 2.4. Functions f and h are C^2 and functions r_i are lsc, finite valued, and prox-bounded.

We remark that prox-boundedness is a general assumption (for example, holds for any function bounded from below) and that we do not assume convexity or weak-convexity (e.g., lower- C^1 , lower- C^2), prox-regularity, or differentiability of r_i .

Let us define the shorthand

$$r_{\lambda}(x) = \sum_{i \in \mathcal{K}} e_{\lambda}r_i(x).$$

Also, we denote the feasible set of the master problem (1.1)–(1.2) (also of (1.4)–(1.5) above) by

$$C := \{x \in \mathbb{R}^n : h(x) = 0, x_{\ell} \leq x \leq x_u\}.$$

For simplicity, we assume that the box bounds x_{ℓ} and x_u are finite so that C is compact. A more general assumption would be that the feasible sublevel sets of F are compact, but this would introduce considerable additional technical complexity without adding much insight.

Assumption 2.5. The constraint set C is nonempty and compact.

3. Moreau continuation: well-posedness and consistency. This section is concerned with the well-posedness of the continuation and proves that the minimizers of the Moreau envelope-regularized problem (1.4)–(1.5) converge, and every cluster point is a minimizer of the original problem (1.1)–(1.2) as $\lambda \downarrow 0$.

Let us define the extended-real objectives

$$G(x) = F(x) + \delta_C(x) \text{ and } G_{\lambda}(x) = F_{\lambda}(x) + \delta_C(x).$$

THEOREM 3.1. Choose $\bar{\lambda} > 0$ below the prox-thresholds of r_i , $i \in \mathcal{K}$, and consider a sequence $\{\lambda_k\}_{k \in \mathbb{N}}$ such that $\lambda_k \downarrow 0$ and $\lambda_k < \bar{\lambda}$. Then the following properties hold under Assumptions 2.4 and 2.5.

- (i) $\{G_{\lambda_k}\}$ converges pointwise and monotonically to G , $G_{\lambda_k}(x) \uparrow G(x)$ for all x .
- (ii) $\{G_{\lambda_k}\}$ epi-converges to G , denoted as $G_{\lambda_k} \xrightarrow{e} G$.
- (iii) Optimal values converge,

$$\inf_{x \in C} F_{\lambda_k}(x) = \inf_x G_{\lambda_k}(x) \rightarrow \inf_x G(x) = \inf_{x \in C} F(x).$$

- (iv) Moreover, for each k , the regularized problem (1.4)–(1.5) admits a global minimizer, and the original problem (1.1)–(1.2) admits a global minimizer. For any choice of minimizers $x(\lambda_k)$ of (1.4)–(1.5), the sequence $\{x(\lambda_k)\}$ is bounded and every cluster point x^* is a minimizer of (1.1)–(1.2). If x^* is the unique solution of (1.1)–(1.2), then $x(\lambda_k) \rightarrow x^*$.

Proof. (i) For $x \notin C$, one has $G_{\lambda_k}(x) = G(x) = \infty$. For $x \in C$, since r_i are prox-bounded proper lsc functions, the Moreau envelope satisfies [30, Theorem 1.25]

$$e_{\lambda_k} r_i(x) \uparrow r_i(x) \quad \text{as } \lambda_k \downarrow 0,$$

which immediately proves $G_{\lambda_k}(x) \uparrow G(x)$.

(ii) The sequence $\{G_{\lambda_k}\}$ is pointwise nondecreasing and each G_{λ_k} is lsc, being the sum of a smooth function, upper- C^2 functions, and the indicator function of the closed set C . The monotone epi-limit equals the pointwise supremum of the lsc functions [30, Proposition 7.4(d)], namely, $G_{\lambda_k} \xrightarrow{e} G$.

(iii) We first remark that the domains of G_{λ_k} are the compact set C for all k and hence are uniformly bounded. This ensures that all $\{G_{\lambda_k}\}$ are eventually level-bounded [30, Exercise 7.32(a)]. Using epi-convergence and eventual level-boundedness, [30, Theorem 7.33] implies that $\inf G_{\lambda_k} \rightarrow \inf G$, which proves (iii).

(iv) Since C is compact and G_{λ_k} and G are lsc with effective domain C , the existence of minimizers for all regularized problems and for the original problem follows from the Weierstrass theorem. Under Assumption 2.5, $\{x(\lambda_k)\}$ is bounded and we consider any cluster point x^* of it. Theorem 7.33 of [30] also implies that x^* is a minimizer of G . If $\operatorname{argmin} G$ is a singleton, then every convergent subsequence has x^* as the cluster point, hence $\{x(\lambda_k)\} \rightarrow x^*$. $\square$

Theorem 3.1 can be viewed as a continuation consistency result, namely, *global* minimizers of the regularized problems converge (along subsequences) to *global* minimizers of the original problem as $\lambda \downarrow 0$. A stronger result holds when the Moreau envelope is over the whole objective, *i.e.*, one uses $e_\lambda(f + \sum_{i \in \mathcal{K}} r_i)$ instead of regularizing each term r_i . Minimizers of the whole-objective regularized problem coincide with minimizers of $f + \sum_{i \in \mathcal{K}} r_i$ for $\lambda > 0$ [30, Example 1.46] under the assumption of prox-boundedness. Furthermore, the local minimizers are also preserved under similar assumptions as recently proved in [19].

Remark 3.2. In our setup, with the Moreau envelope distributed inside the minimization, such results involving local minimizers do not necessarily hold in the nonconvex case as illustrated by the following unconstrained counterexample. Take $f(x) = x^3 + x^4 - \frac{1}{2}x^2$, $r_1(x) = \frac{1}{2}x^2$, and $\mathcal{K} = \{1\}$. Using that $e_\lambda r_1(x) = \frac{x^2}{2(1+\lambda)}$, we see that F_λ has three stationary points: a global minimizer $x_1(\lambda) = \frac{1}{8} \left(-3 - \sqrt{\frac{9+25\lambda}{1+\lambda}} \right)$, a local minimizer $x_2(\lambda) = \frac{1}{8} \left(-3 + \sqrt{\frac{9+25\lambda}{1+\lambda}} \right)$, and a local maximizer 0, for any $\lambda > 0$.

310 As $\lambda \downarrow 0$, $x_1(\lambda)$ converges to $x^* = -\frac{3}{4}$, the global minimizer of F , consistent with
 311 Theorem 3.1; however, the local minimizer $x_2(\lambda)$ collapses to zero, which is only a
 312 stationary point of F .

313 **4. Numerical algorithm.** In this section, we present the continuation algo-
 314 rithm and analyze its convergence properties. We start by describing the algorithm,
 315 which is summarized in listing Algorithm 4.1. In Sections 4.1 and 4.2 we analyze its
 316 convergence properties for the fixed- λ problems (1.4)–(1.5) and of the overall contin-
 317 uation scheme, respectively.

318 The continuation strategy consists of using a sequence of regularization param-
 319 eters $\lambda_s \downarrow 0$ and iteratively solving fixed- λ problems (1.4)–(1.5) to approximate sta-
 320 tionarity. In what follows, for each λ_s the iterations of the algorithm are denoted by
 321 $\tilde{x}^k$, $k = \{0, 1, 2, \dots\}$ and referred to as inner iterations. For better readability, the
 322 iterates corresponding to each λ_s are denoted as x^s and referred to as outer iterates.

323 Consider at some fixed $\lambda_s > 0$ the inner iterate $\tilde{x}^k$. A new inner iteration $\tilde{x}^{k+1}$ is
 324 obtained by means of an inexact stationary point $\tilde{x}^{k+1}$ of a NLP model approximation
 325 of (1.4)–(1.5) in the form

$$326 \quad (4.1) \quad \min_{x \in \mathbb{R}^n} M(x; \tilde{x}^k, g^k; \lambda_s, \alpha_k) := f(x) + r_{\lambda_s}(\tilde{x}^k) + (g^k)^\top (x - \tilde{x}^k) + \alpha_k \|x - \tilde{x}^k\|^2$$

$$327 \quad (4.2) \quad \text{s.t.} \quad h(x) = 0, x_\ell \leq x \leq x_u.$$

328 Similar to the SQP approach of Wang and Petra [33] we use a convex quadratic model
 329 for the nonsmooth objective that uses first-order derivative information g^k defined
 330 below in (4.4) and a quadratic penalty term for step/update stabilization controlled
 331 by the scalar $\alpha_k > 0$. At variance with the SQP approach, we enforce the smooth
 332 constraints (4.2) exactly. This is done for practical considerations since it allows
 333 using off-the-shelf NLP solvers to compute the new iterate, but it also simplifies the
 334 analysis. We believe that the SQP approach [33] can be revisited and used to replace
 335 the NLP approach present here at the cost of increased complexity in the analysis and
 336 implementation since it requires using feasibility restoration to deal with potentially
 337 inconsistent SQP linearizations of the constraints. In contrast, the sequential NLP
 338 approach here leaves this complication to the underlying NLP solver.

339 We do not require that the candidate $\tilde{x}^{k+1}$ be a local or global minimizer of the
 340 NLPs (4.1) – (4.2) since this may be unrealistic. Instead, we only require a stationary
 341 point $\tilde{x}^{k+1}$ of the NLP model that satisfies a weak descent condition

$$342 \quad (4.3) \quad M(\tilde{x}^{k+1}; \tilde{x}^k, g^k; \lambda_s, \alpha_k) \leq M(\tilde{x}^k; \tilde{x}^k, g^k; \lambda_s, \alpha_k).$$

343 We remark that there is always a path in C along which the descent condition holds,
 344 unless $\tilde{x}^k$ is a stationary point. Indeed, if $\tilde{x}^k$ is not a critical point of the NLP
 345 model, then there must exist a direction d in the tangent cone of C to $\tilde{x}^k$ such that
 346 $d^\top \nabla M(\tilde{x}^k) < 0$; otherwise, if every tangent direction d satisfies $d^\top \nabla M(\tilde{x}^k) \geq 0$,
 347 we would have $-\nabla M(\tilde{x}^k) \in N_C(\tilde{x}^k)$, meaning $\tilde{x}^k$ is stationary. Moreover, using
 348 definition of tangents, there is a path $x(\beta) = \tilde{x}^k + \beta d + o(\beta) \in C$, $\beta \in [0, \bar{\beta}]$ such
 349 that $M(x(\beta); \tilde{x}^k, g^k; \lambda, \alpha) = M(\tilde{x}^k; \tilde{x}^k, g^k; \lambda, \alpha) + \beta d^\top \nabla M(\tilde{x}^k) + o(\beta) < M(\tilde{x}^k)$ for all
 350 $\beta \in [0, \bar{\beta}]$ and small enough $\bar{\beta} > 0$. Therefore, the NLP model has a feasible path of
 351 descent and the descent condition (4.3) is a mild requirement for the underlying NLP
 352 solver and less intrusive than requiring a minimizer.

353 Above, g^k encapsulates “derivative” information, namely,

$$354 \quad (4.4) \quad g^k = \sum_{i \in \mathcal{K}} g_i^k \text{ with } g_i^k = \frac{1}{\lambda_s} (\tilde{x}^k - w_i^k) \text{ and } w_i^k \in \text{prox}_{\lambda_s} r_i(\tilde{x}^k).$$

We use slightly less regular and more general subgradients than Wang and Petra [33], namely, we rely on proximal oracle subgradients g_i^k , which are limiting subgradients at “nearby” points (see Lemma 2.1) and are defined for discontinuous functions, while the mentioned previous work employs Clarke subgradients. The proximal oracle subgradients used in g^k above are a subset of the Clarke subgradients of the Moreau envelopes in general, see Theorem 4.6 and its proof of Clarke stationarity. The choice of proximal oracle subgradients g_i^k s (as opposed to choosing g_i^k more restrictively, $g_i^k \in \partial e_{\lambda_s} r_i(\tilde{x}^k)$) stems from the representations of subdifferentials of Moreau envelopes of general functions satisfying Assumption 2.4, as discussed after Lemma 2.1.

We use an accept-reject rule based on a majorization-ratio test that updates the penalty parameter $\alpha^k > 0$ in the NLP model solved at each iteration. Upon solving the NLP model (4.1)–(4.2), we define the predicted and actual reductions at the candidate $\bar{x}^{k+1}$ as

$$(4.5) \quad \begin{aligned} \text{Pred}_k(\bar{x}^{k+1}) &= M(\tilde{x}^k; \tilde{x}^k, g^k; \lambda, \alpha_k) - M(\bar{x}^{k+1}; \tilde{x}^k, g^k; \lambda, \alpha_k) \text{ and} \\ \text{Ared}_k(\bar{x}^{k+1}) &= F_\lambda(\tilde{x}^k) - F_\lambda(\bar{x}^{k+1}). \end{aligned}$$

We define the ratio

$$\rho_k = \frac{\text{Ared}_k(\bar{x}^{k+1})}{\text{Pred}_k(\bar{x}^{k+1}) + \sigma \|\bar{x}^{k+1} - \tilde{x}^k\|^2}.$$

This is analogous to trust-region acceptance, but specialized to our exact constraints model. The iterate $\tilde{x}^{k+1} = \bar{x}^{k+1}$ is accepted whenever $\rho_k \geq \bar{\rho}$ with $\bar{\rho} \in (0, 1)$; otherwise, the step is rejected and α_k increases by a factor of two.

At each outer iteration s , the inner iterates $\tilde{x}^k$ of Algorithm 4.1 (steps 5 – 20) aim to reach an approximate stationary point for (1.4)–(1.5). Inexactness is controlled by the tolerance ε_s and has the goal to balance inner solve accuracy with the continuation error. The stopping criterion of the inner iterates use a lagged stationarity measure in the form of

$$\hat{R}(\tilde{x}^k) := \nabla f(\tilde{x}^k) + g^{k-1} + \nabla h(\tilde{x}^k)^\top \nu^k - \gamma^k + \theta^k,$$

where $\nu^k \in \mathbb{R}^m$, $\gamma^k \geq 0$, and $\theta^k \geq 0$ are the KKT multipliers corresponding to the optimality conditions (see (4.11)–(4.13) for the exact form) of the model NLPs (4.1) – (4.2) under the general MFCQ for C . From these KKT conditions, we remark that $\hat{R}(\tilde{x}^k) = -2\alpha_{k-1}(\tilde{x}^k - \tilde{x}^{k-1})$, which makes this lagged stationarity both computationally cheap and practical (since it does not require dual information from the numerical solver used for the master problem). Therefore, the stopping criterion used for the inner loop is

$$(4.6) \quad \|\hat{R}(\tilde{x}^k)\|_\infty = 2\alpha_{k-1}\|\tilde{x}^k - \tilde{x}^{k-1}\|_\infty \leq \varepsilon_s$$

Later, in Proposition 4.5 we show that the inner loop reaches (4.6) for $\varepsilon_s > 0$ in a finite number of iterations. Theorem 4.6 analyzes the infinite inner loop (without the stopping criterion) and shows that the inner iterates $\tilde{x}^k$ are bounded and any of their accumulation or cluster points $x^{*\lambda}$ are stationary points of (1.4)–(1.5).

Algorithm 4.1 repeats the sequential NLP steps discussed above for a given sequence $\lambda_s \downarrow 0$. It is shown in Theorem 4.11 that under mild conditions the outer iterates x^s of Algorithm 4.1 are bounded and any of their accumulation points x^*

395 satisfy stationarity conditions

$$396 \quad (4.7) \quad 0 \in \nabla f(x^*) + \sum_{i \in \mathcal{K}} \partial r_i(x^*) + \nabla h(x^*)^\top \nu^* - \gamma^* + \theta^*,$$

$$397 \quad (4.8) \quad h(x^*) = 0, \quad 0 \leq \gamma^* \perp (x^* - \ell) \geq 0, \quad 0 \leq \theta^* \perp (u - x^*) \geq 0,$$

398 for some multipliers $\nu^* \in \mathbb{R}^m$, $\gamma^* \geq 0$, and $\theta^* \geq 0$ under MFCQ for the feasible set C .
 399 The limiting stationarity condition (4.7) targeted here is analogous in form to standard
 400 first-order necessary optimality conditions, see, for example, [30]. Consequently, we
 401 use the following basic CQ (BCQ) specific to non-Lipschitz functions, counterpart of
 402 the horizon BCQ from [30, Theorem 8.15].

403 **BCQ for non-Lipschitz objectives.** If there exist $v_i \in \partial^\infty r_i(x^*)$, $i \in \mathcal{K}$, such that
 404 $-\sum_{i \in \mathcal{K}} v_i \in N_C(x^*)$, then $v_i = 0$ for all $i \in \mathcal{K}$. Under MFCQ, the BCQ is equivalent
 405 to: if there exist $v_i \in \partial^\infty r_i(x^*)$, $i \in \mathcal{K}$, and $\nu \in \mathbb{R}^m$, $\gamma \geq 0$, and $\theta \geq 0$ satisfying
 406 complementarity conditions from (4.8) such that $\sum_{i \in \mathcal{K}} v_i + \nabla h(x^*)^\top \nu - \gamma + \theta = 0$,
 407 then $v_i = 0$ for all $i \in \mathcal{K}$.

Algorithm 4.1 Proximal oracle sequential NLP continuation algorithm

```

1: Initial point  $x^0 \in C$  and  $\lambda_0$  below the prox-thresholds of all  $r_i$ ; parameters  $\bar{\rho} \in$ 
   (0, 1],  $\sigma > 0$ , and  $\underline{\alpha} > 0$ ; sequences of tolerances  $\varepsilon_s \downarrow 0$  and regularizations  $\lambda_s \downarrow 0$ .
2:  $s \leftarrow 0$ 
3: while true do
4:   Inner loop initializations:  $k \leftarrow 0$ ,  $\tilde{x}^0 \leftarrow x^s$ ,  $\alpha_k \leftarrow \max\{\underline{\alpha}, \frac{|K|}{2\lambda_s}\}$ 
5:   while true do
6:     for  $i \in \mathcal{K}$  do
7:       Compute  $w_i^k \in \text{prox}_{\lambda_s} r_i(\tilde{x}^k)$  and  $g_i^k = (\tilde{x}^k - w_i^k)/\lambda_s$ 
8:     end for
9:      $g^k \leftarrow \sum_{i \in \mathcal{K}} g_i^k$ 
10:    Compute a stationary point  $\bar{x}^{k+1}$  of  $M(\cdot; \tilde{x}^k, g^k; \lambda_s, \alpha_k)$  from (4.1)–(4.2) sat-
       isfying (4.3)
11:    Compute  $\rho_k(\bar{x}^{k+1}) = \text{Ared}_k(\bar{x}^{k+1})/(\text{Pred}_k(\bar{x}^{k+1}) + \sigma \|\bar{x}^{k+1} - \tilde{x}^k\|^2)$  using (4.5)
       if  $\bar{x}^{k+1} \neq \tilde{x}^k$ , otherwise set  $\rho_k(\bar{x}^{k+1}) = 1$ 
12:    if  $\rho_k(\bar{x}^{k+1}) \geq \bar{\rho}$  then
13:       $\tilde{x}^{k+1} \leftarrow \bar{x}^{k+1}$ ,  $\alpha_{k+1} \leftarrow \alpha_k$ ,  $k \leftarrow k + 1$  {accept iterate}
14:    else
15:       $\tilde{x}^{k+1} \leftarrow \tilde{x}^k$ ,  $\alpha_{k+1} \leftarrow 2\alpha_k$  {reject iterate}
16:       $g^{k+1} \leftarrow g^k$ ,  $g_i^{k+1} \leftarrow g_i^k$ ,  $w_i^{k+1} \leftarrow w_i^k$ ,  $i \in \mathcal{K}$ ,  $k \leftarrow k + 1$ 
17:      Return to step 10
18:    end if
19:    if  $\|\hat{R}(\tilde{x}^k)\|_\infty \leq \varepsilon_s$  then
20:      break {stopping criterion for inner loop}
21:    end if
22:  end while
23:   $s \leftarrow s + 1$ ,  $x^s \leftarrow \tilde{x}^k$ ,  $g_i^s \leftarrow g_i^{k-1}$ ,  $w_i^s \leftarrow w_i^{k-1}$  for all  $i \in \mathcal{K}$ 
24: end while

```

408 BCQ rules out abnormal stationarity conditions and is a prerequisite for nec-
 409 essary first-order optimality conditions (4.7)–(4.8) to hold for constrained problems
 410 with nonsmooth and nonconvex objectives. In this respect, it can be viewed as the
 411 counterpart of KKT conditions, which rule out Fritz-John points with zero objective

multipliers, in smooth constrained optimization. Geometrically, near x^* , limiting subgradients can become arbitrarily large, and their normalized limiting directions belong to the horizon subgradients $\partial^\infty r_i(x^*)$ (for some $i \in \mathcal{K}$). If one such direction exists and is opposite to the normal cone of C at some boundary point, or, equivalently, cancels out the multiplier terms in the BCQ above, then the constraint boundary can absorb this unbounded tilt. In such cases general first-order stationarity conditions such as [30, Theorem 8.15] may fail at minimizers as shown in Examples 4.1 and 4.2. The BCQ excludes exactly this pathology. We remark that the BCQ holds automatically for locally Lipschitz functions r_i , since $\partial^\infty r_i(x^*) = \{0\}$.

Example 4.1. Consider $\min_x \{r_1(x) := x^{1/3} \mid x \geq 0\}$ with the (global) minimizer $x^* = 0$. This problem fails BCQ above since there exists $v = \frac{1}{3} \in \partial^\infty r_1(0)$ such that $v - \gamma = 0$, where $0 \leq x \perp \gamma \geq 0$. We remark that $\partial r_1(0) = \emptyset$ and therefore the necessary first-order stationary conditions (4.7)–(4.8) for $f = 0$ or ones in [30] (Theorem 8.15) do not hold at this minimizer.

Example 4.2. The following example shows that (global) minimizers fail to satisfy first-order stationarity when the BCQ does not hold even when the subdifferential is nonempty and the feasible set satisfies MFCQ. Consider

$$\min_{x \in \mathbb{R}^2} \{r_1(x_1, x_2) := x_1^2 + x_1 + \sqrt{\max(x_2, 0)} \mid x \in C\},$$

with $C := \{x \in \mathbb{R}^2 : h(x) := x_1^2 - x_2 \leq 0\}$. Since $r_1(x_1, x_2) = x_1^2 + x_1 + \sqrt{x_2} \geq x_1^2 + x_1 + |x_1|$ over $x = (x_1, x_2) \in C$, we see that $x^* = (0, 0)$ is a strict local/global minimizer over C . We remark that MFCQ holds at x^* and that $\partial r_1(0, 0) = \hat{\partial} r(0, 0) = \{1\} \times \mathbb{R}_+$.

BCQ fails at x^* since $0 \neq v = (0, 1) \in \partial^\infty r(0, 0)$ satisfies $v + \nu(0, -1) = 0$ with $0 \leq \nu \perp -h(0, 0) \geq 0$ (for $\nu = 1$). Here we remark that ν is signed since $h(x)$ is involved in an inequality constraint and, also, the bound multipliers γ and θ are not present. To see that $(0, 1) \in \partial^\infty r_1(0, 0)$, take $x_\tau = (0, \tau)$ with $\tau \downarrow 0$. As $x_\tau \rightarrow (0, 0)$, we have $r_1(x_\tau) = \sqrt{\tau} \rightarrow 0$ and $v^\tau := \nabla r(0, \tau) = (1, \frac{1}{2\sqrt{\tau}}) \in \hat{\partial} r_1(x_\tau)$. By choosing $t_\tau = 2\sqrt{\tau} \rightarrow 0$ we obtain $t_\tau v^\tau = (2\sqrt{\tau}, 1) \rightarrow (0, 1)$. Then $(0, 1) \in \partial^\infty r_1(0, 0)$ by definition of horizon subdifferential from Section 2. In fact, one can prove that $\partial^\infty r_1(0, 0) = \{0\} \times \mathbb{R}_+$.

Even though $(0, 0)$ is a minimizer the stationarity conditions do not hold since

$$0 \notin \partial r_1(0, 0) + \nu(0, -1) = \{1\} \times \mathbb{R}_+ + \nu(0, -1)$$

for any ν such that $0 \leq \nu \perp h(0, 0) \geq 0$.

We use the concept of sum-stationarity using $\sum_{i \in \mathcal{K}} \partial r_i$, which comes naturally in our composite objective, instead of $\partial(\sum_{i \in \mathcal{K}} r_i)$ commonly used with weakly convex summands. To the best of our knowledge, no inclusion is guaranteed between the two sets in our general setup without additional qualifications. We show in Lemma 4.12 that $\sum_{i \in \mathcal{K}} \partial r_i \subseteq \partial(\sum_{i \in \mathcal{K}} r_i)$ for subdifferentially regular functions, which cover weakly convex functions, hence the stationarity conditions used in this work characterize the typical limiting stationarity for this class of problems.

4.1. Inner-loop convergence. We start with a majorization lemma specific to upper- C^2 functions [33]. Here we provide a stronger version for Moreau envelopes with an explicit bound in terms of the regularization λ and proximal oracle subgradients.

LEMMA 4.3. *For any $\lambda > 0$ below the prox-thresholds of $r_i(\cdot)$ and a given $\bar{x}$, the*

455 following majorization bound holds: for any $\bar{g}_i \in \frac{1}{\lambda}(\bar{x} - \text{prox}_\lambda r_i(\bar{x}))$, $i \in \mathcal{K}$,

$$456 \quad r_\lambda(x) \leq r_\lambda(\bar{x}) + \langle \bar{g}, x - \bar{x} \rangle + Q_\lambda \|x - \bar{x}\|^2, \quad \forall x, \quad \text{where } \bar{g} = \sum_{i \in \mathcal{K}} \bar{g}_i \text{ and } Q_\lambda = \frac{|\mathcal{K}|}{2\lambda}.$$

457 *Proof.* Fix $i \in \mathcal{K}$ and consider $\bar{g}_i \in \frac{1}{\lambda}(\bar{x} - \text{prox}_\lambda r_i(\bar{x}))$, namely, there exists $w_i \in$
 458 $\text{prox}_\lambda r_i(\bar{x})$ such that $\bar{g}_i = \frac{1}{\lambda}(\bar{x} - w_i)$. The optimality of w_i with respect to the
 459 minimization problem defining the Moreau envelope implies $e_{\lambda_i} r_i(x) \leq r_i(w_i) + \frac{1}{2\lambda} \|x -$
 460 $w_i\|^2$ for all x . Also, for $x = \bar{x}$ we have $e_{\lambda_i} r_i(\bar{x}) = r_i(w_i) + \frac{1}{2\lambda} \|\bar{x} - w_i\|^2$. By subtracting
 461 these two conditions, we obtain after some manipulations of the quadratic terms that

$$462 \quad e_{\lambda_i} r_i(x) \leq e_{\lambda_i} r_i(\bar{x}) + \frac{1}{2\lambda} \langle x - \bar{x} + \bar{x} - w_i, x - \bar{x} + \bar{x} - w_i \rangle - \frac{1}{2\lambda} \|\bar{x} - w_i\|^2$$

$$463 \quad = e_{\lambda_i} r_i(\bar{x}) + \langle \frac{1}{\lambda}(\bar{x} - w_i), x - \bar{x} \rangle + \frac{1}{2\lambda} \|x - \bar{x}\|^2. \quad \square$$

464 Finally, summing over $i \in \mathcal{K}$ gives the desired majorization bound.

465 The above lemma is the key to proving the convergence of the proposed algorithm.
 466 First, we use it to prove that the candidate iterates $\bar{x}^{k+1}$ are eventually accepted.

467 **LEMMA 4.4.** *Fix an outer iteration s . Then only finitely many inner steps can be*
 468 *rejected in Algorithm 4.1. Consequently, the penalty parameters satisfy $\alpha_k \in [\underline{\alpha}, \bar{\alpha}_s]$,*
 469 *with $\bar{\alpha}_s = \max\{\underline{\alpha}, 2(Q_\lambda + \bar{\rho}\sigma)\}$, for all k greater than some iteration threshold index.*

470 *Proof.* We first show that whenever α_k becomes larger than $Q_\lambda + \bar{\rho}\sigma$, all sub-
 471 sequent iterations are accepted. Assume that at iterate k , the trial point $\bar{x}^{k+1}$ is
 472 computed with $\alpha_k > Q_\lambda + \bar{\rho}\sigma$. We use the majorization of Lemma (4.3) to write

$$473 \quad F_\lambda(\bar{x}^{k+1}) = f(\bar{x}^{k+1}) + r_\lambda(\bar{x}^{k+1}) \leq f(\bar{x}^{k+1}) + r_\lambda(\tilde{x}^k) + \langle g^k, \bar{x}^{k+1} - \tilde{x}^k \rangle +$$

$$474 \quad Q_\lambda \|\bar{x}^{k+1} - \tilde{x}^k\|^2 = M_k(\bar{x}^{k+1}; \tilde{x}^k, g^k; \lambda, \alpha_k) + (Q_\lambda - \alpha_k) \|\bar{x}^{k+1} - \tilde{x}^k\|^2.$$

475 By subtracting the above from $F_\lambda(\tilde{x}^k) = M_k(\tilde{x}^k; \tilde{x}^k, g^k; \lambda, \alpha_k)$ and using our working
 476 assumption that $\alpha_k > Q_\lambda + \bar{\rho}\sigma$, we obtain

$$477 \quad \text{Ared}_k(\bar{x}^{k+1}) \geq \text{Pred}_k(\bar{x}^{k+1}) + (\alpha_k - Q_\lambda) \|\bar{x}^{k+1} - \tilde{x}^k\|^2$$

$$478 \quad > \text{Pred}_k(\bar{x}^{k+1}) + \bar{\rho}\sigma \|\bar{x}^{k+1} - \tilde{x}^k\|^2 \geq \bar{\rho}(\text{Pred}_k(\bar{x}^{k+1}) + \sigma \|\bar{x}^{k+1} - \tilde{x}^k\|^2), \blacksquare$$

479 with the last inequality due to the algorithmic parameter $\bar{\rho} \leq 1$ and $\text{Pred}_k(\bar{x}^{k+1}) \geq 0$.
 480 Therefore, $\rho_k(\bar{x}^{k+1}) \geq \bar{\rho}$ and $\bar{x}^{k+1}$ will be accepted in the acceptance test from step 13
 481 of Algorithm 4.1, namely $\tilde{x}^{k+1} = \bar{x}^{k+1}$, and, also, $\alpha_{k+1} = \alpha_k$. Proceeding by induction
 482 through the next iterations, it should be apparent that all iterations $k + j$ with $j \geq 1$
 483 of Algorithm 4.1 accept $\bar{x}^{k+j}$ and $\alpha_{k+j} = \alpha_k$ for all $j \geq 1$.

484 We remark that the inner loop of Algorithm 4.1 may satisfy the termination
 485 criteria in step 20 and exit after finite number of iterations without increasing the
 486 penalty parameters α_k above the $Q_\lambda + \bar{\rho}\sigma$ threshold. In this case, it should be apparent
 487 that $\alpha_k \in [\underline{\alpha}, Q_\lambda + \bar{\rho}\sigma]$ for all k . If this is not the case, let k_0 denote the smallest
 488 iteration index such that $\alpha_{k_0} > Q_\lambda + \bar{\rho}\sigma$. This index must be finite because of the
 489 geometric increase of penalties α_k for the rejected iterates.

490 If $k_0 = 0$, then $\alpha_{k_0} = \max\{\underline{\alpha}, Q_\lambda\} \in [\underline{\alpha}, \max\{\underline{\alpha}, Q_\lambda\}]$ due to initialization. If
 491 $k_0 \geq 1$, then we immediately see that the previous step must have been rejected,
 492 therefore, $\alpha_{k_0-1} = \frac{1}{2}\alpha_{k_0} < Q_\lambda + \bar{\rho}\sigma$, which implies that $\alpha_{k_0} < 2(Q_\lambda + \bar{\rho}\sigma)$. Finally,

we remark that $\alpha_{k_0} = \alpha_{k_0+j} = \dots < 2(Q_\lambda + \bar{\rho}\sigma)$ for all $j \geq 1$, as per the first part of the proof.

The expression for $\bar{\alpha}_s$ follows after collecting the lower and upper bounds for α_k over the above three cases. $\square$

PROPOSITION 4.5. *At any outer iteration s , the inner iterates $\{\tilde{x}^k\}$ of Algorithm 4.1 are bounded and*

(i) *$\{F_\lambda(\tilde{x}^k)\}$ is decreasing monotonically:*

$$F_\lambda(\tilde{x}^{k+1}) - F_\lambda(\tilde{x}^k) \leq -\bar{\rho}\sigma\|\tilde{x}^{k+1} - \tilde{x}^k\|^2, \forall k \geq 0;$$

(ii) *the updates $\Delta^k = \tilde{x}^{k+1} - \tilde{x}^k$ are square-summable, that is, $\sum_k \|\Delta_k\|^2 < \infty$, and, thus, $\Delta_k \rightarrow 0$;*

(iii) *for any tolerance $\varepsilon_s > 0$, the inner-loop stopping criterion (4.6) is reached in a finite number of inner iterations k_s .*

Proof. We first remark that the sequence $\{\tilde{x}^k\}$ is bounded since it is generated by the algorithm in the feasible set C , which is compact by the Assumption 2.5.

(i) By Lemma 4.4 all iterates k greater than some index k_0 are accepted in step 13 of the algorithm, hence

$$F_\lambda(\tilde{x}^k) - F_\lambda(\tilde{x}^{k+1}) \geq \bar{\rho}(\text{Pred}_k(\tilde{x}^{k+1}) + \sigma\|\tilde{x}^{k+1} - \tilde{x}^k\|^2) \geq \bar{\rho}\sigma\|\tilde{x}^{k+1} - \tilde{x}^k\|^2.$$

The iterates 0 to k_0 also satisfy the monotonicity: the accepted iterates in the form above, while the rejected iterates trivially, since $\tilde{x}^{k+1} = \tilde{x}^k$.

(ii) Let us sum over $k \in [k_0, \bar{k}]$ the inequality from (i):

$$\bar{\rho}\sigma \sum_{k=k_0}^{\bar{k}} \|\Delta_k\|^2 \leq F_\lambda(\tilde{x}^{k_0}) - F_\lambda(\tilde{x}^{\bar{k}}).$$

Since $f(\cdot)$ is C^2 and the Moreau envelopes defining $r_\lambda(\cdot)$ are upper- C^2 , hence Lipschitz continuous, the function $F_\lambda(\cdot)$ is continuous. Therefore, boundedness of $\{\tilde{x}^k\}$ implies that $\{F_\lambda(\tilde{x}^k)\}$ is bounded. Taking the limit $\bar{k} \rightarrow \infty$ we obtain that $\bar{\rho}\sigma \sum_k \|\Delta_k\|^2 < \infty$. This completes the proof of (ii) since $\bar{\rho}\sigma > 0$.

(iii) First observe that $\|\hat{R}(\tilde{x}^k)\| = 2\alpha_k\|\tilde{x}^k - \tilde{x}^{k-1}\| \leq 2\bar{\alpha}_s\|\Delta_{k-1}\|$ by Lemma 4.4. Using $\Delta_k \rightarrow 0$ from (ii), stopping criterion (4.6) is going to be met for sufficiently large k , so the inner loop terminates after finitely many iterations. $\square$

The convergence behavior of the inner loop (for fixed λ) of Algorithm 4.1 fits under a standard subsequence convergence and only requires a standard CQ on the feasible set of the target problem. We assume MFCQ.

THEOREM 4.6. *Assume MFCQ holds for the feasible set C , in addition to Assumptions 2.4 and 2.5. Fix an outer iteration s and consider the sequence of inner iterates $\{\tilde{x}^k\}$ generated by the inner loop of of Algorithm 4.1 without the stopping test in step 20, i.e., the infinite loop consisting of steps 5–17. Then $\{\tilde{x}^k\}$ is bounded and all its cluster points $x^{*\lambda}$ satisfy the proximal limiting stationarity conditions*

$$(4.9) \quad 0 = \nabla f(x^{*\lambda}) + \sum_{i \in \mathcal{K}} g_i^{*\lambda} + \nabla h(x^{*\lambda})^\top \nu^{*\lambda} - \gamma^{*\lambda} + \theta^{*\lambda},$$

$$(4.10) \quad h(x^{*\lambda}) = 0, \quad 0 \leq \gamma^{*\lambda} \perp (x^{*\lambda} - \ell) \geq 0, \quad 0 \leq \theta^{*\lambda} \perp (u - x^{*\lambda}) \geq 0.$$

for some multipliers $\nu^{*\lambda} \in \mathbb{R}^m$, $\gamma^{*\lambda} \geq 0$, and $\theta^{*\lambda} \geq 0$ and with $g_i^{*\lambda} = \frac{1}{\lambda}(x^{*\lambda} - w_i^{*\lambda}) \in \partial r_i(w_i^{*\lambda})$, where $w_i^{*\lambda} \in \text{prox}_\lambda r_i(x^{*\lambda})$.

Moreover, $x^{*\lambda}$ also satisfies Clarke stationarity conditions

$$0 \in \nabla f(x^{*\lambda}) + \partial^C r_\lambda(x^{*\lambda}) + \nabla h(x^{*\lambda})^\top \nu^{*\lambda} - \gamma^{*\lambda} + \theta^{*\lambda},$$

$$h(x^{*\lambda}) = 0, \quad 0 \leq \gamma^{*\lambda} \perp (x^{*\lambda} - \ell) \geq 0, \quad 0 \leq \theta^{*\lambda} \perp (u - x^{*\lambda}) \geq 0.$$

for some multipliers $\nu^{*\lambda}$, $\gamma^{*\lambda} \geq 0$, and $\theta^{*\lambda} \geq 0$.

Proof. The boundedness of $\{\tilde{x}^k\}$ from Proposition 4.5 shows that cluster points of $\{\tilde{x}^k\}$ exist. Let $x^{*\lambda}$ be an arbitrary such point and, without re-indexing, let $\tilde{x}^k \rightarrow x^{*\lambda}$.

By Lemma 4.4, $\tilde{x}^{k+1}$ becomes an accepted iterate for large enough k and therefore satisfies the KKT conditions for the sequential NLP model (4.1)-(4.2), namely, there exist multipliers ν^{k+1} , γ^{k+1} , and θ^{k+1} such that

$$(4.11) \quad \nabla f(\tilde{x}^{k+1}) + g^k + 2\alpha_k \Delta^k + \nabla h(\tilde{x}^{k+1})^\top \nu^{k+1} - \gamma^{k+1} + \theta^{k+1} = 0,$$

$$(4.12) \quad h(\tilde{x}^{k+1}) = 0,$$

$$(4.13) \quad 0 \leq \gamma^{k+1} \perp (\tilde{x}^{k+1} - \ell) \geq 0, \quad 0 \leq \theta^{k+1} \perp (u - \tilde{x}^{k+1}) \geq 0.$$

The main steps of the proof of limiting stationarity equations (4.9)-(4.10) are:

- (a) show the sequence $\{g^k\}_k$ is bounded and its cluster points $g^{\lambda*}$ satisfy $g^{\lambda*} = \sum_{i \in \mathcal{K}} g_i^{\lambda*}$ with $g_i^{\lambda*} = \frac{1}{\lambda}(x^{\lambda*} - w_i^{\lambda*}) \in \partial r_i(w_i^{\lambda*})$ and $w_i^{\lambda*} \in \text{prox}_\lambda r_i(x^{\lambda*})$, for all $i \in \mathcal{K}$.
- (b) show the sequences $\{\nu^{k+1}\}_k$, $\{\gamma^{k+1}\}_k$, and $\{\theta^{k+1}\}_k$ of multipliers are bounded as well;
- (c) use (a) and (b) to take the limit $k \rightarrow \infty$ in the above KKT system to obtain the conclusion.

To prove (a), we first look at each g_i^k term from the expression of g^k in steps 7 and 9 of Algorithm 4.1 (see also (4.4)). We use a well-known proximal behavior, namely, if $\tilde{x}_k \rightarrow x^{\lambda*}$ and $w_i^k \in \text{prox}_\lambda r_i(\tilde{x}_k)$, then the sequence $\{w_i^k\}$ is bounded and its cluster points $w_i^{\lambda*}$ lie in $\text{prox}_\lambda r_i(x^{\lambda*})$ [30, Proposition 1.25]. Hence the sequence of $g_i^k = \frac{1}{\lambda}(\tilde{x}_k - w_i^k)$ is bounded, and so is the sequence of their sum, which is g^k . Similarly, cluster points satisfy the relationship from (i), with the observation that the relationship $g_i^{\lambda*} \in \partial r_i(w_i^{\lambda*})$ follows from (i) of Lemma 2.1.

We prove (b) by contradiction, namely assume there exists an unbounded subsequence (not relabeled) such that $t_k := \|(\nu^k, \gamma^k, \theta^k)\| \rightarrow \infty$. Since $\|(\frac{\nu^k}{t_k}, \frac{\gamma^k}{t_k}, \frac{\theta^k}{t_k})\| = 1$ for all k , there must exist a cluster point and a subsequence (not relabeled) such that

$$(4.14) \quad \left(\frac{\nu^k}{t_k}, \frac{\gamma^k}{t_k}, \frac{\theta^k}{t_k} \right) \rightarrow (\hat{\nu}, \hat{\gamma}, \hat{\theta}) \text{ as } k \rightarrow \infty \text{ with } \|(\hat{\nu}, \hat{\gamma}, \hat{\theta})\| = 1.$$

Next consider (4.11) rescaled by t_{k+1} , namely

$$\frac{1}{t_{k+1}} (\nabla f(\tilde{x}^{k+1}) + g^k + 2\alpha_k \Delta^k) + \nabla h(\tilde{x}^{k+1})^\top \frac{\nu^{k+1}}{t_{k+1}} - \frac{\gamma^{k+1}}{t_{k+1}} + \frac{\theta^{k+1}}{t_{k+1}} = 0.$$

Recall that we work with a convergent (sub)sequence $\{\tilde{x}^{k+1}\} \rightarrow x^{\lambda*}$, which makes the term $\nabla f(\tilde{x}^{k+1}) + g^k + 2\alpha_k \Delta^k$ bounded/convergent due to ∇f being continuous (by Assumption 2.4), $\{g_k\}$ being bounded (by (i) above), $\Delta^k \rightarrow 0$ (by Proposition 4.5), and $\alpha_k \leq \bar{\alpha}_s < \infty$ (by Lemma 4.4). Also $\nabla h(\tilde{x}^{k+1}) \rightarrow \nabla h(x^{\lambda*})$ since ∇h is continuous per Assumption 2.4. We see that by also using (4.14), we can take the limit to obtain

$$\nabla h(x^{\lambda*})^\top \hat{\nu} - \hat{\gamma} + \hat{\theta} = 0.$$

Similar continuity arguments can be applied to (4.13) to obtain that

$$0 \leq \hat{\gamma} \perp (x^{\lambda^*} - \ell) \geq 0, \quad 0 \leq \hat{\theta} \perp (u - x^{\lambda^*}) \geq 0.$$

By MFCQ, the only multipliers $\hat{\nu}, \hat{\gamma}, \hat{\theta}$ satisfying the above relationships are the trivial ones, $(\hat{\nu}, \hat{\gamma}, \hat{\theta}) = (0, 0, 0)$, which contradicts (4.14). Therefore, $\{\nu^{k+1}\}_k$, $\{\gamma^{k+1}\}_k$, and $\{\theta^{k+1}\}_k$ are bounded.

We now prove (c). First consider any cluster point $(\nu^{\lambda^*}, \gamma^{\lambda^*}, \theta^{\lambda^*})$ of $(\nu^k, \gamma^k, \theta^k)$ and cluster point g^{λ^*} of g^k . Since $\Delta_k \rightarrow 0$ by Proposition 4.5, $\alpha_k \leq \bar{\alpha}_s \leq \infty$ by Lemma 4.4, and $\nabla f, \nabla h$, and h are continuous by Assumption 2.4, we take the limit over any converging subsequence in (4.11)–(4.13) to obtain (4.9)–(4.10).

The Clarke stationarity follows from (4.9)–(4.10) since, as we next show, $\sum_{i \in \mathcal{K}} g_i^{\lambda^*} \in \partial^C r_\lambda(x^{\lambda^*})$. A general characterization [18][Proposition 3.1 and Theorem 3.2] for the Clarke subdifferentials of Moreau envelopes $e_\lambda r_i(\cdot)$ is available via convex hulls involving proximal points, namely,

$$\partial^C e_\lambda r_i(x) = \text{co} \left\{ \frac{1}{\lambda}(x - w) : w \in \text{prox}_\lambda r_i(x) \right\}.$$

Therefore, $g_i^{\lambda^*}$ defined in (4.9)–(4.10) are Clarke subgradients of the Moreau envelopes,

$$g_i^{\lambda^*} \in \partial^C e_\lambda f_i(x^{\lambda^*}), \quad \forall i \in \mathcal{K}.$$

By Lemma A.1, we have that $\sum_{i \in \mathcal{K}} g_i^{\lambda^*} \in \partial^C (\sum_{i \in \mathcal{K}} e_\lambda r_i(x^{\lambda^*})) = \partial^C r_\lambda(x^{\lambda^*})$, which concludes the proof. $\square$

Remark 4.7. Clarke stationarity conditions (ii) of Theorem 4.6 are weaker than the (limiting) sum-stationarity from (i) since the Clarke subdifferential is the convex hull of the limiting subdifferential in our general setup. This is a major improvement over our previous algorithm for upper- C^2 [33], mainly possible because we exploit the additional structure that Moreau envelopes have over general upper- C^2 functions, namely, the relationships between proximal points (minimizers of the Moreau envelope) and limiting subdifferentials from Lemma 2.1. In this respect, (4.9)–(4.10) of Theorem 4.6 can be viewed as stationarity conditions involving limiting subgradients at “nearby” points. More important, they are instrumental in proving the (subsequence) convergence of the outer iterates in the limit of Moreau parameter $\lambda \downarrow 0$. Clarke stationarity conditions (ii) are too weak to obtain the same convergence result.

4.2. Outer-loop convergence. In this section we elucidate the convergence behavior of the outer loop of Algorithm 4.1. We start with a technical lemma showing that proximal points of convergent sequences are also convergent.

LEMMA 4.8. *Consider a sequence x^s that converges to x^* and a sequence $\lambda_s \downarrow 0$. For any choice of proximal points $w_i^s \in \text{prox}_{\lambda_s} r_i(x^s)$, the sequences $\{w_i^s\}$ converge to x^* for all $i \in \mathcal{K}$.*

Proof. The proof is common to all $i \in \mathcal{K}$, therefore let $r := r_i$ and $w_s := w_i^s$. Prox-boundedness of r implies that there exists a fixed $\bar{\lambda} > 0$ such that $\bar{m} := e_{\bar{\lambda}}(x^*) > -\infty$. We then use the definition (1.3) of the Moreau envelope to write $r(w^s) \geq \bar{m} - \frac{1}{2\bar{\lambda}} \|w^s - x^*\|^2$. By the choice of $w^s \in \text{prox}_{\lambda_s} r(x^s)$, for $\lambda_s < \bar{\lambda}/2$, we can write

$$(4.15) \quad r(w^s) + \frac{1}{2\lambda_s} \|w^s - x^s\|^2 \leq r(x^*) + \frac{1}{2\lambda_s} \|x^s - x^*\|^2.$$

613 The previous two inequalities imply that

$$614 \quad \bar{m} - \frac{1}{2\bar{\lambda}} \|w^s - x^*\|^2 + \frac{1}{2\lambda_s} \|w^s - x^s\|^2 \leq r(x^*) + \frac{1}{2\lambda_s} \|x^s - x^*\|^2.$$

615 Now use $\|w^s - x^s\|^2 = \|(w^s - x^*) - (x^s - x^*)\|^2 \geq \frac{1}{2} \|w^s - x^*\|^2 - \|x^s - x^*\|^2$ and bring
616 the terms involving $\|x^s - x^*\|^2$ to the left to obtain

$$617 \quad \bar{m} + \left(\frac{1}{4\lambda_s} - \frac{1}{2\bar{\lambda}} \right) \|w^s - x^*\|^2 \leq r(x^*) + \frac{1}{\lambda_s} \|x^s - x^*\|^2.$$

618 Consequently, $\|w^s - x^*\|^2 \leq \frac{4\bar{\lambda}}{\bar{\lambda} - 2\lambda_s} [\lambda_s(r(x^*) - \bar{m}) + \|x^s - x^*\|^2]$, which proves $w^s \rightarrow$
619 x^* . $\square$

620 One would expect the function values $r_i(w_i^s) \rightarrow r_i(x^*)$ as well; however, this is
621 not necessarily the case when $r_i(\cdot)$ has a jump discontinuity at x^* . For example,
622 $r(x) = 1$ for $x \neq 0$ and $r(0) = 0$, the Moreau envelope is $e_\lambda(x) = \min\{1, \frac{|x|^2}{2\lambda}\}$; also,
623 $\text{prox}_\lambda r(x) = \{0\}$ for $|x| < \sqrt{2\lambda}$, $\text{prox}_\lambda r(x) = \{0, x\}$ for $|x| = \sqrt{2\lambda}$, and $\text{prox}_\lambda r(x) =$
624 $\{x\}$ for $|x| > \sqrt{2\lambda}$. For simplicity, let us work with only one term, $\mathcal{K} = \{1\}$, and
625 set $r_i(x) = r(x)$. Assume that Algorithm 4.1 generates $x^s = \sqrt{2\lambda_s}$ and that the
626 proximal oracle always returns $w_i^s = \sqrt{2\lambda_s} \in \text{prox}_{\lambda_s} r(x) = \{0, \sqrt{2\lambda_s}\}$. We observe
627 that $r_i(w_i^s) \rightarrow 1 \neq r_i(0)$, even though $w_i^s \rightarrow x^* = 0$. This pathology has previously
628 been recognized in [5], where an attentive convergence condition is used (see their
629 assumption (H3)) to prove convergence of their PPA framework, namely the condition
630 below.

631 *Condition 4.9.* For convergent sequences $x^s \rightarrow x^*$ and corresponding $w_i^s \in \text{prox}_{\lambda_s} r_i(x^s)$,
632 attentive convergence holds along $\{w_i^s\}$, namely $r_i(w_i^s) \rightarrow r_i(x^*)$, for all $i \in \mathcal{K}$. $\blacksquare$

633 This condition is quite general and is satisfied, for example, when the functions
634 $r_i(\cdot)$ are continuous. It should also be noted that it holds in the lsc case whenever
635 the outer iterates x^s , as returned by the inner loop of Algorithm 4.1, are exact min-
636 imizers of the regularized problem (1.4)–(1.5) as $\lambda_s \downarrow 0$. The latter fact is proved in
637 Lemma 4.15.

638 Given two tolerances $\varepsilon > 0$ and $\delta > 0$, a point $x \in C$ is (ε, δ) -approximately sum-
639 stationary if there exist points w_i , subgradients $g_i \in \partial r_i(w_i)$, and a normal vector
640 $\eta \in N_C(x)$ such that

$$641 \quad \|w_i - x\|_\infty \leq \delta, \quad \forall i \in \mathcal{K}, \quad \text{and} \quad \|\nabla f(x) + \sum_{i \in \mathcal{K}} g_i + \eta\|_\infty \leq \varepsilon.$$

642 In words, $x \in C$ is (ε, δ) -approximately sum-stationary if it is stationary up to a
643 residual norm ε with subgradients sampled at points no further than δ from x .

644 Our main and general convergence results are provided by the following proposi-
645 tion. In the following, let k_s be the number of inner iterations at each outer iteration
646 s in Algorithm 4.1. We recall that $x^s = \tilde{x}^{k_s}$ according to our notation convention;
647 also, let $x_-^s = \tilde{x}^{k_s-1}$ and $\Delta^s = x^s - x_-^s$.

648 **PROPOSITION 4.10.** *Consider the iterates sequence $\{x^s\}$ generated by Algorithm 4.1*
649 *under a sequence of regularization parameters $\lambda_s \downarrow 0$ and inner-loop tolerances $\varepsilon_s \downarrow 0$.*
650 *Assume that Condition 4.9 holds along subsequences $x^s \rightarrow x^*$, that the conditions of*
651 *Theorem 4.6 hold, that BCQ holds at cluster points x^* , and that Assumptions 2.4*
652 *and 2.5 are satisfied. Then:*

(i) Each outer-loop iterate x^s satisfies

$$\|\nabla f(x^s) + \sum_{i \in \mathcal{K}} g_i^s + \nabla h(x^s)^\top \nu^s - \gamma^s + \theta^s\|_\infty \leq \varepsilon_s,$$

together with the KKT feasibility $x^s \in C$ and KKT complementarity conditions. As a result x^s is $(\varepsilon_s, \delta_s)$ -approximately sum-stationary, with

$$\delta_s := \max_{i \in \mathcal{K}} \|x^s - w_i^s\|_\infty \leq \frac{\varepsilon_s}{2\alpha} + \lambda_s \max_{i \in \mathcal{K}} \|g_i^s\|_\infty.$$

(ii) The sequences $\{g_i^s\}$ and $\{\eta^s\}$, where $\eta^s := \nabla h(x^s)^\top \nu^s - \gamma^s + \theta^s$, are bounded. Consequently, $\delta_s \leq \frac{\varepsilon_s}{2\alpha} + M_g \lambda_s$ with $M_g := \sup_s \max_{i \in \mathcal{K}} \|g_i^s\|_\infty < \infty$ and, therefore $\delta_s \downarrow 0$.

(iii) For any prescribed tolerances $\lambda_{tol} > 0$ and $\varepsilon_{tol} > 0$, Algorithm 4.1 with stopping criteria $\lambda_s \leq \lambda_{tol}$ and $\varepsilon_s \leq \varepsilon_{tol}$ terminates after finitely many iterations and returns a point $\bar{x}$ that is $(\varepsilon_{\bar{s}}, \delta_{\bar{s}})$ -approximately sum-stationary, with $\delta_{\bar{s}} \leq \frac{\varepsilon_{tol}}{2\alpha} + M_g \lambda_{tol}$.

Proof. (i) Fix an outer iteration s . At the terminating inner iteration index k_s , which is finite per Proposition 4.5, the KKT conditions of the model NLP at this iteration (see (4.11)–(4.13)) give

$$\nabla f(x^s) + \sum_{i \in \mathcal{K}} g_i^s + \eta^s = -2\alpha_{k_s-1}(x^s - x_-^s),$$

where η^s is the shorthand defined in (ii) above. We also remark that $\hat{R}(x^s) = \nabla f(x^s) + \sum_{i \in \mathcal{K}} g_i^s + \eta^s$ simply by the form of η^s and definition of $\hat{R}(x^s)$ (see above (4.6)). By the algorithm's inner loop stopping criterion (4.6) at iteration k_s , we have that $\|\hat{R}(x^s)\|_\infty \leq \varepsilon_s$, yielding the stationarity inequality from (i). The feasibility and complementarity conditions follow directly from the aforementioned KKT conditions of the model NLP.

The inner loop stopping criterion $\|\hat{R}(x^s)\|_\infty \leq \varepsilon_s$ also implies

$$\|x^s - x_-^s\|_\infty \leq \frac{\varepsilon_s}{2\alpha_{k_s-1}} \leq \frac{\varepsilon_s}{2\alpha},$$

where the last inequality is due to the lower bound for penalty parameters α_k from Lemma 4.4.

On the other hand, since $x_-^s - w_i^s = \lambda_s g_i^s$ for all $i \in \mathcal{K}$, we can write

$$\|x^s - w_i^s\|_\infty \leq \|x^s - x_-^s\|_\infty + \|x_-^s - w_i^s\|_\infty = \|x^s - x_-^s\|_\infty + \lambda_s \|g_i^s\|_\infty,$$

which proves the $(\varepsilon_s, \delta_s)$ -approximate sum-stationarity of x^s .

(ii) Since $\{x^s\} \in C$, Assumption 2.5 ensures $\{x^s\}$ is bounded. Let

$$e^s = \nabla f(x^s) + \sum_{i \in \mathcal{K}} g_i^s + \eta^s.$$

Suppose by contradiction that the family $\{g_i^s\}_s$, $i \in \mathcal{K}$ and $\{\eta^s\}_s$ is unbounded as $\lambda_s \rightarrow 0$. Passing to an unbounded subsequence and relabeling if necessary, we may assume $\sum_{i \in \mathcal{K}} \|g_i^s\| + \|\eta^s\| \rightarrow +\infty$. Define the scaling

$$t_s = \frac{1}{1 + \sum_{i \in \mathcal{K}} \|g_i^s\| + \|\eta^s\|} \downarrow 0,$$

688 and multiply stationarity identity above by t_s to obtain

$$689 \quad t_s \nabla f(x^s) + \sum_{i \in \mathcal{K}} t_s g_i^s + t_s \eta^s = t_s e^s.$$

690 Since x^s is bounded and $\nabla f(\cdot)$ is continuous, $\{\nabla f(x^s)\}$ must be bounded, hence
 691 $t_s \nabla f(x^s) \rightarrow 0$. Similarly, since $\|e^s\|_\infty = \|\hat{R}(x^s)\|_\infty \leq \varepsilon_s$, we must have $t_s e^s \rightarrow 0$.
 692 Hence,

$$693 \quad (4.16) \quad \sum_{i \in \mathcal{K}} t_s g_i^s + t_s \eta^s \rightarrow 0.$$

694 On the other hand, simply by the definition of t_s and the unboundedness hypoth-
 695 esis, we observe that

$$696 \quad \sum_{i \in \mathcal{K}} \|t_s g_i^s\| + \|t_s \eta^s\| = \frac{\sum_{i \in \mathcal{K}} \|g_i^s\| + \|\eta^s\|}{1 + \sum_{i \in \mathcal{K}} \|g_i^s\| + \|\eta^s\|} \rightarrow 1,$$

697 and, therefore, $\{(t_s g_i^s)_{i \in \mathcal{K}}, t_s \eta^s\}$ has a nonzero cluster point. Passing to a subsequence
 698 if needed, we obtain

$$699 \quad t_s g_i^s \rightarrow v_i^\infty \quad \forall i \in \mathcal{K} \text{ and } t_s \eta^s \rightarrow \eta^\infty,$$

700 with at least one of the v_i^∞ and η^∞ being nonzero.

701 Consider a convergent subsequence, still not relabeled, such that $x^s \rightarrow x^*$. From
 702 part (i), $\|x^s - x_-^s\| \leq \frac{\varepsilon_s}{2\alpha}$, hence $x_-^s \rightarrow x^*$ as well. Lemma 4.8 applied to $\{x_-^s\}$
 703 implies $w_i^s \rightarrow x^*$; also $r_i(w_i^s) \rightarrow r_i(x^*)$ under the r_i -attentive convergence of $\{w_i^s\}$
 704 from Condition 4.9.

705 We next prove $v_i^\infty \in \partial^\infty r_i(x^*)$. We first pass to regular subgradients required by
 706 the definition of horizon subdifferentials. Since $g_i^s \in \partial r_i(w_i^s)$, definition of the limiting
 707 subdifferential yields points $\hat{w}_i^s$ and $\hat{g}_i^s \in \hat{\partial} r_i(\hat{w}_i^s)$ for all s such that

$$708 \quad \|\hat{w}_i^s - w_i^s\| \leq t_s, \quad |r_i(\hat{w}_i^s) - r_i(w_i^s)| \leq t_s, \quad \text{and} \quad \|\hat{g}_i^s - g_i^s\| \leq t_s.$$

709 Then $\hat{w}_i^s \rightarrow x^*$, $r_i(\hat{w}_i^s) \rightarrow r_i(x^*)$, and

$$710 \quad \|t_s \hat{g}_i^s - v_i^\infty\| \leq t_s \|\hat{g}_i^s - g_i^s\| + \|t_s g_i^s - v_i^\infty\| \rightarrow 0.$$

711 With these regular subgradients, the definition of the horizon subdifferential provides

$$712 \quad v_i^\infty \in \partial^\infty r_i(x^*), \quad \forall i \in \mathcal{K}.$$

713 Similarly, since each $N_C(x^s)$ is a cone and $\eta^s \in N_C(x^s)$, we have $t_s \eta^s \in N_C(x^s)$.
 714 By outer semicontinuity of the limiting normal cone [30, Proposition 6.6], $x^s \rightarrow x^*$
 715 and $t_s \eta^s \rightarrow \eta^\infty$ imply $\eta^\infty \in N_C(x^*)$. Taking the limit $t_s \downarrow 0$ in (4.16), we obtain
 716 $\sum_{i \in \mathcal{K}} v_i^\infty + \eta^\infty = 0$. BCQ at x^* then forces $v_i^\infty = 0$ for all i and $\eta^\infty = 0$, contradicting
 717 that at least one cluster is nonzero. Therefore $\{g_i^s\}$ and $\{\eta^s\}_s$ are bounded.

718 (iii) Since $\lambda_s \downarrow 0$ and $\varepsilon_s \downarrow 0$, there exists a finite outer index $\bar{s}$ with $\lambda_{\bar{s}} \leq \lambda_{tol}$
 719 and $\varepsilon_{\bar{s}} \leq \varepsilon_{tol}$. By Proposition 4.5 for all outer iterations $s \leq \bar{s}$ the inner iterations
 720 terminate finitely, therefore x^s is reached in a finite number of iterations. The $(\varepsilon_{\bar{s}}, \delta_{\bar{s}})$ -
 721 approximate stationarity follows from (i) and (ii) above. $\square$

722 An explicit coupling between inner-loop tolerance ε_s and regularization λ_s from
 723 Proposition 4.10 should be used by a numerical method based on Algorithm 4.1 to

properly balance accuracy of the inner and outer solves. The key bound is given by Proposition 4.10:

$$\delta_s \leq \frac{\varepsilon_s}{2\alpha} + \lambda_s \max_{i \in \mathcal{K}} \|g_i^s\|_\infty = \frac{\varepsilon_s}{2\alpha} + \max_{i \in \mathcal{K}} \|x_-^s - w_i^s\|_\infty.$$

A natural choice is to set ε_s the same order as the proximal displacement $B_s := \max_{i \in \mathcal{K}} \|x_-^s - w_i^s\|_\infty$. Otherwise, if $\varepsilon_s \ll B_s$, the regularized subproblem (1.4)–(1.5) is oversolved during the inner iterations; and, if $B_s \ll \varepsilon_s$, the overall error is dominated by the inner-loop accuracy. A practical, a posteriori rule for the inner-loop termination (step 20 in the algorithm) would therefore be

$$\|\hat{R}(\tilde{x}^k)\|_\infty \leq B \max_{i \in \mathcal{K}} \|\tilde{x}^k - w_i^k\|_\infty \text{ or, equivalently, } \|\hat{R}(\tilde{x}^k)\|_\infty \leq B \lambda_s \max_{i \in \mathcal{K}} \|g_i^k\|_\infty,$$

for some prescribed $B > 0$.

THEOREM 4.11. *Consider Algorithm 4.1 under a sequence of regularizations $\lambda_s \downarrow 0$ and inner-loop tolerances $\varepsilon_s \downarrow 0$. Under the same assumptions as in Proposition 4.10, any cluster point x^* of the bounded sequence $\{x^s\}$ satisfies the sum-stationarity conditions (4.7)–(4.8) for the target problem (1.1)–(1.2).*

Proof. The sequence $\{x^s\}$ is bounded since C is compact by Assumption 2.5. We take an arbitrary accumulation point x^* and a convergent subsequence (not relabeled) $x^s \rightarrow x^*$. By (ii) of Proposition 4.10, we can take subsequences, not relabeled, such that

$$g_i^s \rightarrow g_i^*, \forall i \in \mathcal{K}, \text{ and } \eta^s \rightarrow \eta^*.$$

Possibly after a relabeling, for $x^s \rightarrow x^*$ we have $\nabla f(x^s) \rightarrow \nabla f(x^*)$ by continuity of ∇f given by Assumption 2.4. Also by closedness of normal cones $\eta^* \in N_C(x^*)$.

Using a similar argument as in the proof of (ii) of Proposition 4.10, one can prove that $x_-^s \rightarrow x^*$ as well. Lemma 4.8 applied to the sequence $\{x_-^s\}$ implies $w_i^s \rightarrow x^*$. Since $g_i^s \in \partial r_i(w_i^s)$ and $g_i^s \rightarrow g_i^*$, under the r_i -attentive convergence of $\{w_i^s\}$ from Condition 4.9, we must have $g_i^* \in \partial r_i(x^*)$ for each $i \in \mathcal{K}$, by the outer semicontinuity of limiting subdifferential [30, Proposition 8.7].

With the above convergence properties, we can take the limit of the stationarity condition from (i) of Proposition 4.10 to obtain

$$\nabla f(x^*) + \sum_{i \in \mathcal{K}} g_i^* + \eta^* = 0, \text{ with } g_i^* \in \partial r_i(x^*) \text{ and } \eta^* \in N_C(x^*).$$

Under the MFCQ for the feasible set C of the master problem (see Theorem 4.6), $\eta^* \in N_C(x^*)$ has the representation $\eta^* = \nabla h(x^*)^\top \nu^* - \gamma^* + \theta^*$ with the multipliers ν^* , γ^* , and θ^* satisfying sign/complementarity conditions from (4.8). Finally, the stationarity equation above implies the stationarity condition (4.7), which completes the proof. $\square$

A sharper iteration-complexity analysis via Kurdyka–Łojasiewicz (KL) assumption and arguments is not immediate in our framework, since our stationarity measure is based on proximal oracle subgradients, whereas the standard KL machinery is typically formulated in terms of the limiting subdifferential and its associated slope. Relating these two notions in a way that yields clean rate complexity estimates would very likely require additional regularity assumptions, so we defer such a KL-based analysis to future work.

4.3. Stronger convergence guarantees under additional regularity. For several classes of objective functions r_i covered by Assumption 2.4, including lower- C^1 , lower- C^2 , and (lsc) convex functions, the sum-stationarity obtained above can be strengthened to the more familiar limiting subgradient stationarity. Moreover, the convergence prerequisites from the previous section hold automatically for broader classes of nonsmooth objectives, including locally Lipschitz functions (which encompass all of the aforementioned classes) and semismooth functions.

LEMMA 4.12. *If the functions r_i satisfying Assumption 2.4 are also subdifferentially regular in the sense of Rockafellar and Wets [30, Definition 7.25] (also known as lower regular [24]), the sum-stationarity condition (4.7) from Theorem 4.11 takes the more commonly used form*

$$0 \in \nabla f(x^*) + \partial \left(\sum_{i \in \mathcal{K}} r_i(x^*) \right) + \nabla h(x^*)^\top \nu^* - \gamma^* + \theta^*.$$

Proof. The stationarity condition follows from

$$\sum_{i \in \mathcal{K}} \partial r_i(x^*) = \sum_{i \in \mathcal{K}} \hat{\partial} r_i(x^*) \subseteq \hat{\partial} \left(\sum_{i \in \mathcal{K}} r_i(x^*) \right) \subseteq \partial \left(\sum_{i \in \mathcal{K}} r_i(x^*) \right),$$

where the first inclusion is given by the sum rule for regular subgradients [30, Corollary 10.9], the second inclusion is due to the regular subdifferential being a subset of the limiting subdifferential, and the equality is due to r_i being subdifferentially regular. $\square$

We remark that if r_i are also locally Lipschitz, then the horizon qualification from Corollary 10.9 of [30] implies the stronger relationship $\sum_{i \in \mathcal{K}} \partial r_i(x^*) = \partial \left(\sum_{i \in \mathcal{K}} r_i(x^*) \right)$. $\blacksquare$

Under the Lipschitzian property for objective terms r_i , the conditions needed for the finite termination and subsequence convergence of Proposition 4.10 and Theorem 4.11 can be considerably relaxed, as shown in the following proposition.

PROPOSITION 4.13. *If the functions r_i are locally Lipschitz, the finite convergence properties of Proposition 4.10 and the asymptotic convergence of Theorem 4.11 hold only under Assumptions 2.4 and 2.5 and MFCQ of Theorem 4.6 since Condition 4.9 and BCQ hold automatically.*

Proof. The proof is immediate once we observe that (i) $\partial^\infty r_i(\bar{x}) = \{0\}$ by local Lipschitz continuity [25], hence BCQ is satisfied, and (ii) Condition 4.9 also holds due to continuity of r_i . $\square$

Remark 4.14. Proposition 4.13 applies to semismooth functions [22], lower- C^1 , lower- C^2 , and (lsc) convex functions, which are locally Lipschitz.

The following lemma shows that Condition 4.9 holds for the general case (r_i only satisfy Assumption 2.4 and are possibly discontinuous) if the inner iterations of Algorithm 4.1 finish with (approximate) minimizers x^s of the inner, fixed- λ problem (1.4)–(1.5) instead of (approximate) stationary points (in the sense of (i) of Proposition 4.10). This occurs, for example, when the feasible set C and the objective f in the master problem are convex and it may also occur as well when a global numerical solver is used for the inner problem.

LEMMA 4.15. *Consider Algorithm 4.1 under a sequence of regularization parameters $\lambda_s \downarrow 0$. Suppose that, for each outer iteration s , the inner loop terminates after a finite number k_s of inner iterations and returns the outer iterate $x^s := \tilde{x}^{k_s} \in C$,*

806 *satisfying*

$$807 \quad F_{\lambda_s}(x^s) \leq \inf\{F_{\lambda_s}(x) : x \in C\} + \tau_s,$$

808 *for some sequence $\tau_s \downarrow 0$. Let $x^s \rightarrow x^*$ be a convergent subsequence of such outer*
 809 *iterates. Then Condition 4.9 holds along this subsequence.*

810 *Proof.* Let $w_i^s \in \text{prox}_{\lambda_s} r_i(x^s)$ and define

$$811 \quad t_i^s := \frac{1}{2\lambda_s} \|x^s - w_i^s\|^2 \geq 0.$$

812 From the choice of w_i^s , we have $e_{\lambda_s} r_i(x^s) = r_i(w_i^s) + t_i^s$, and therefore $F_{\lambda_s}(x^s) =$
 813 $f(x^s) + \sum_{i \in K} (r_i(w_i^s) + t_i^s)$.

814 Since $x^* \in C$ and x^s is τ_s -suboptimal for F_{λ_s} over C ,

$$815 \quad F_{\lambda_s}(x^s) \leq \inf\{F_{\lambda_s}(x) : x \in C\} + \tau_s \leq F_{\lambda_s}(x^*) + \tau_s.$$

816 Also, $e_{\lambda_s} r_i(x^*) \leq r_i(x^*)$ by the monotonicity of the Moreau envelope (see Theo-
 817 rem 3.1), hence

$$818 \quad f(x^s) + \sum_{i \in K} (r_i(w_i^s) + t_i^s) \leq f(x^*) + \sum_{i \in K} r_i(x^*) + \tau_s.$$

819 Since f is continuous, $x^s \rightarrow x^*$, and $\tau_s \rightarrow 0$, we obtain

$$820 \quad (4.17) \quad \limsup_{s \rightarrow \infty} \sum_{i \in K} (r_i(w_i^s) + t_i^s) \leq \sum_{i \in K} r_i(x^*).$$

821 On the other hand, Lemma 4.8 gives $w_i^s \rightarrow x^*$, and lower semicontinuity of r_i implies

$$822 \quad r_i(x^*) \leq \liminf_{s \rightarrow \infty} r_i(w_i^s), \quad \forall i \in K.$$

823 Summing over $i \in K$ and using $t_i^s \geq 0$, we get

$$824 \quad \sum_{i \in K} r_i(x^*) \leq \sum_{i \in K} \liminf_{s \rightarrow \infty} r_i(w_i^s) \leq \liminf_{s \rightarrow \infty} \sum_{i \in K} r_i(w_i^s) \leq \liminf_{s \rightarrow \infty} \sum_{i \in K} (r_i(w_i^s) + t_i^s),$$

825 where the second inequality uses properties of $\liminf$ (i.e., Fatou's Lemma). Combin-
 826 ing this with (4.17) yields

$$827 \quad \lim_{s \rightarrow \infty} \sum_{i \in K} (r_i(w_i^s) + t_i^s) = \sum_{i \in K} r_i(x^*).$$

828 Finally, for each $i \in K$, the lower semicontinuity of r_i , nonnegativity of t_i^s , and
 829 $w_i^s \rightarrow x^*$ imply

$$830 \quad \liminf_{s \rightarrow \infty} (r_i(w_i^s) + t_i^s) \geq \liminf_{s \rightarrow \infty} r_i(w_i^s) \geq r_i(x^*).$$

831 The last two relations imply $r_i(w_i^s) + t_i^s \rightarrow r_i(x^*)$, $\forall i \in K$. This and $w_i^s \rightarrow x^*$ imply

$$832 \quad 0 \leq \limsup_{s \rightarrow \infty} t_i^s = \limsup_{s \rightarrow \infty} [r_i(w_i^s) + t_i^s - r_i(w_i^s)] \leq \limsup_{s \rightarrow \infty} [r_i(w_i^s) + t_i^s] - \liminf_{s \rightarrow \infty} r_i(w_i^s) \\ 833 \quad = r_i(x^*) - \liminf_{s \rightarrow \infty} r_i(w_i^s) \leq 0,$$

834 therefore $t_i^s \rightarrow 0$ and, consequently, $r_i(w_i^s) \rightarrow r_i(x^*)$, proving Condition 4.9. $\square$

5. A case study in parametric optimization with applications to electrical power grids. As mentioned in the introduction, a main motivation for the present framework comes from optimization models in which the nonsmooth terms r_i are scenario-wise optimal value functions. A representative form is

$$(5.1) \quad \begin{aligned} r_i(x) &:= \min_{y_1, y_2} c_i(y_1, y_2) \\ \text{s.t. } x - \Delta_i &\leq y_1 \leq x + \Delta_i, \quad y_\ell \leq y_2 \leq y_u, \quad h_i(y_1, y_2) = 0, \end{aligned}$$

where x denotes the master decision, y_1 collects scenario variables constrained to remain close to x , y_2 collects scenario-local state variables, $\Delta_i > 0$ prescribes admissible post-scenario deviations, and $i \in K$ indexes scenarios, *e.g.*, system contingencies or uncertainty realizations. We use this bound-coupling form only for notational simplicity. The same discussion applies when x also enters the objective or the general constraints of the scenario problem.

This is precisely the setting in which Algorithm 4.1 is useful. In many applications, evaluating $r_i(x)$ already requires solving a large nonconvex NLP. The main difficulty is therefore not the function evaluation itself, but obtaining reliable first-order information for the value function r_i . Even when the scenario data are C^2 , the marginal function r_i is, in general, only lsc [30]. Under additional constraint qualifications such as MFCQ, it becomes locally Lipschitz [13, 14]. However, even under MFCQ, both Clarke and limiting subdifferentials may fail to admit multiplier-based representations in degenerate or unstable nonconvex cases, see for example, Example 1 in [13] and Theorem 4 in [23], respectively. Algorithm 4.1 avoids differentiating through the scenario NLP (5.1). Instead of relying on potentially numerically unavailable subgradients, it requires only a proximal point of r_i , which can be obtained by solving essentially the same scenario problem with an additional convex quadratic term. Concretely, one needs to solve

$$\begin{aligned} e_\lambda r_i(x) &= \min_{y_1, y_2, w} c_i(y_1, y_2) + \frac{1}{2\lambda} \|w - x\|^2 \\ \text{s.t. } w - \Delta_i &\leq y_1 \leq w + \Delta_i, \quad y_\ell \leq y_2 \leq y_u, \quad h_i(y_1, y_2) = 0, \end{aligned}$$

and any minimizer w_i^λ yields a proximal oracle subgradient $g_i^\lambda = \frac{1}{\lambda}(x - w_i^\lambda)$. Thus, once a solver for the scenario NLP is available, the oracle required by Algorithm 4.1 can often be implemented with minimal structural changes, without differentiating through KKT multipliers or relying on sensitivity routines.

This viewpoint is particularly natural for preventive security-constrained optimal power flow models (SC-OPF) and related stochastic programming models relevant in power dispatch operations under stochastic renewable generation, loads, and operating conditions [3]. The master variable x represents base-case power dispatch and control settings. Each scenario $i \in K$ corresponds to an $N - 1$ contingency, such as the loss of a generator, line, or transformer. For fixed x , the scenario subproblem (5.1) computes a post-contingency AC operating point subject to AC power flow equations, nodal active/reactive balance, generator capability limits, emergency voltage and thermal limits, and contingency transition rules that describe the post-contingency response and grid state via the constraints in (5.1). The scenario objective c_i is often convex, as it captures post-contingency penalties such as line-overload penalties and active or reactive power imbalance penalties; it may include other monetary costs associated with post-contingency response.

Algorithm 4.1 matches the operational structure of such models in several ways. First, the master constraints (1.2) are enforced exactly, so every accepted iterate is a

feasible base-case operating point. This is especially attractive in real-time grid operations, which are subject to stringent time-to-solution requirements, where preserving feasibility and security is often more important than aggressively pursuing marginal cost decreases. Second, the proximal oracles are scenario-separable: for a given x , each r_i or $e_\lambda r_i$ can be evaluated independently, so the contingency solves can be carried out termwise and, when available, in parallel. Third, the method returns first-order information in a form that is immediately usable by the master algorithm, namely the proximal oracle subgradients g_i^λ , without requiring the delicate post-processing of dual information from the inner NLP solves. In [27] we successfully employed derivative-free convex proximal stabilizations for r_i , constructed using power systems engineering insight, to address the pervasive difficulties associated with derivative evaluations of r_i . This strategy, however, entails a loss of robustness and limits the method’s applicability to other classes of optimization problems. In this sense, Algorithm 4.1 turns the proximal stabilization intuition behind decomposition-based SC-OPF heuristics from [27] into a principled continuation method with explicit stationarity guarantees.

The general regularity assumptions and results of Section 4 apply in this setting. In addition to being lsc, the value functions r_i are proper and finite-valued whenever the scenario problems are feasible for all relevant x ; furthermore, since the scenario objective c_i from (5.1) is generally bounded from below, r_i is also prox-bounded and therefore falls under Assumption 2.4. When MFCQ holds for the scenario NLP (5.1), r_i is locally Lipschitz [14], so the stronger Proposition 4.13 applies (since Condition 4.9 and BCQ hold automatically). Thus, the parametric NLP setting provides a concrete class of applications in which the proximal oracle assumptions are natural, verifiable, and directly compatible with the convergence guarantees presented above.

6. Conclusions and future work. We introduced a proximal oracle sequential NLP framework for nonsmooth constrained optimization problems in which the nonsmooth objective is a finite sum of proper lower semicontinuous nonconvex terms accessed through proximal oracles. The method applies Moreau regularization termwise to preserve rich decomposition properties of the original problem and builds exact-constraint NLP models from the proximal points $w_i \in \text{prox}_\lambda r_i(x)$ and the associated proximal oracle subgradients $(x - w_i)/\lambda$. This construction avoids differentiating through scenario subproblems and does not rely on multiplier sensitivity calculations. We proved consistency of the distributed Moreau-envelope continuation at the level of global minimizers. For fixed λ , the inner sequential NLP iteration has cluster points satisfying proximal limiting stationarity and Clarke stationarity for the regularized problem. Along the outer continuation, cluster points satisfy first-order sum-stationarity for the original problem under the stated attentive-convergence and CQ assumptions; when the terms r_i are locally Lipschitz, these additional convergence prerequisites hold automatically. The iterative algorithm terminates finitely for prescribed positive regularization and stationarity tolerances with an explicit error bound. We also show how the required proximal oracles can be computed with minimal modifications of the scenario NLP for the cases when r_i are optimal value functions of parametric problems.

Several directions remain for future work. A primary direction is the development of a robust implementation of the proposed algorithm, including treatment of inexact proximal oracle solves, adaptive choices of regularization and inner-loop tolerances, warm-starting strategies, and safeguards for nonconvex NLP oracles. Such an implementation would enable systematic numerical evaluation on large-scale stochastic programming and SC-OPF instances and would clarify the tradeoffs among

the various algorithmic and computational choices that affect scalability, robustness, and applicability. Further work will also investigate sharper complexity guarantees, for example under a KL assumption. This would require relating the proximal oracle subgradient stationarity used here to a limiting subdifferential slope [5], possibly under additional structure such as prox-regularity [28] or a KL property [4, 5].

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Appendix A. Sum rule for Clarke subdifferentials. The following lemma establishes that the Clarke subdifferential satisfies the exact sum rule for upper- C^2 functions. We include a proof because we were not able to find it established previously in the literature.

LEMMA A.1. *Let $g_1, g_2 : \mathbb{R}^n \rightarrow \mathbb{R}$ be upper- C^2 in a neighborhood of x . Then*

$$\partial^C(g_1 + g_2)(x) = \partial^C g_1(x) + \partial^C g_2(x).$$

Proof. Since g_1 and g_2 are upper- C^2 near x , by the local representation theorem for upper- C^2 functions [30, Theorem 10.33], there exist a neighborhood U of x , C^2 functions $\phi_i : U \rightarrow \mathbb{R}$, and convex functions $q_i : U \rightarrow \mathbb{R}$ such that $g_i = \phi_i - q_i$ on U , $i = \{1, 2\}$. Therefore $g_1 + g_2 = (\phi_1 + \phi_2) - (q_1 + q_2)$ on U , with $\phi_1 + \phi_2$ being C^2 and $q_1 + q_2$ being convex.

For any proper convex function q , the Clarke subdifferential coincides with the convex subdifferential [7], that is, $\partial^C q(x) = \partial q(x)$. Moreover, for a C^1 function ϕ and a locally Lipschitz function ψ , the Clarke sum rule [7] gives $\partial^C(\phi + \psi)(x) = \nabla \phi(x) + \partial^C \psi(x)$. Applying this with $\psi = -q$ and using $\partial^C(-q)(x) = -\partial^C q(x)$, we obtain $\partial^C(\phi - q)(x) = \nabla \phi(x) - \partial q(x)$. Therefore,

$$\partial^C(g_1 + g_2)(x) = \nabla(\phi_1 + \phi_2)(x) - \partial(q_1 + q_2)(x).$$

Since q_1 and q_2 are convex and finite on U , the convex subdifferential sum rule [30] yields $\partial(q_1 + q_2)(x) = \partial q_1(x) + \partial q_2(x)$. Consequently,

$$\begin{aligned} \partial^C(g_1 + g_2)(x) &= \nabla \phi_1(x) + \nabla \phi_2(x) - (\partial q_1(x) + \partial q_2(x)) \\ &= (\nabla \phi_1(x) - \partial q_1(x)) + (\nabla \phi_2(x) - \partial q_2(x)) = \partial^C g_1(x) + \partial^C g_2(x). \end{aligned}$$

This proves the result. $\square$

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